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# Points to Consider in Remote and Hybrid GMP/GDP Inspections

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# Introduction

The COVID-19 pandemic has driven the world to find new technological tools to do business safely without sacrificing quality, efficiency, and effectiveness. This is especially true for the biopharmaceutical industry. While many consider on-site inspections to be the optimal way of assessing compliance, remote inspection tools increasingly are seen as useful ways of obtaining meaningful insights into a site's operations. But what factors should companies and health authorities consider when adjusting to the inspection format selected? How do you determine, address, and plan for the technological hurdles and considerations?

This document examines advantages and best practices for planning and implementing all types of remote inspections, including desktop, virtual, and hybrid inspections. It describes best practices that will help all participants engage efficiently and avoid unnecessary or unexpected delays, stressors, or complications in the remote inspection process. While good planning and preparation by both the inspectors[1] and the inspected entity are important for any inspection, they are vital to an efficient and effective remote inspection. This document therefore presents the topics in order, from planning and notification through closing and response, as illustrated in **Figure 1**. The suggestions in this document may be useful to sites that are the target of inspections and to health authorities.

If inspections conducted using remote tools can provide thorough and reliable information, health authorities may continue to use them in certain situations in the future. Remote tools present advantages, as they allow health authorities to make optimal use of resources based on risk, allow for the involvement of specialized experts in a way that has not traditionally been possible on-site, and are more environmentally friendly. Issues currently seen as constraints may become less problematic as sites and health authorities gain experience.

In addition, **Appendix 2** presents the complete results of a PDA global survey on biopharmaceutical manufacturing experiences related to remote inspections**.** These results may provide insights about industry's needs and concerns.

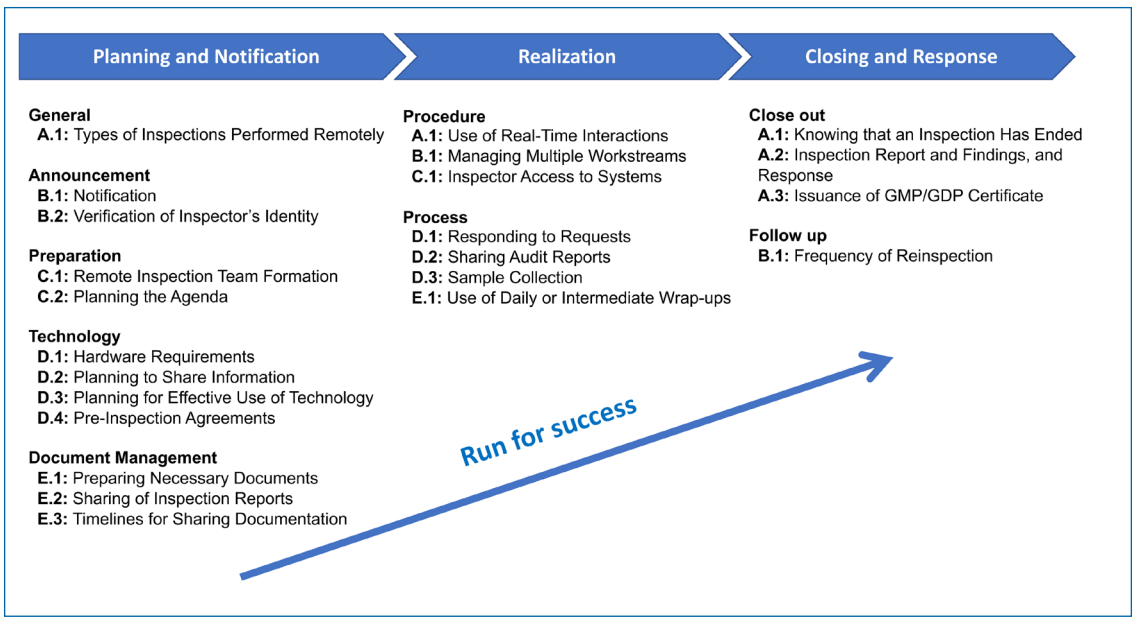


**Figure 1** Flow of the Sections of this Points to Consider Document

1 This document uses the term "Inspector" to refer to individuals representing a health authority during an inspection. Some health authorities may use different terms.

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Because health authorities around the world may take different approaches or be governed by specific legal requirements, sites should always consider local laws and practices. As with all PDA technical documents addressing new and evolving processes, PDA may update or adjust these Points to Consider as more experience is gained. This document does not represent a standard or regulatory guidance.

## Scope

This document applies only to regulatory inspections concerning good manufacturing practices (GMP) and good distribution practices (GDP) and provides information relevant to both health authorities and the industry. The recommendations described in this document may be applied to other types of GxP inspections, although those were not specifically considered in the development of this document.

This document is not intended to address remote auditing programs, such as remote supplier audits, vendor audits, internal audits, or self-inspections, although similar challenges apply.

## Glossary

This document uses the following definitions. Because of the diversity of terminology used by inspectorates around the world, however, sites should be aware that inspectors may use a different term or definition than stated here.

**Desktop Inspection:** A type of remote inspection in which documentation (e.g., filing commitments, site description, and key procedures) is requested by the inspecting body that is not physically present at the site(s). Requests and responses to questions are shared in written form (e.g., via email or on a document-sharing platform). There can be several iterations of the requests and review of documentation, questions, and answers. Phone calls and videoconferences take place only rarely to provide clarification about the documents requested or provided.[2]

**Document-Sharing Platform**: A secure software application that facilitates file transfer. Document-sharing platforms allow end users to upload files to a shared storage space and designate who may access the files.

**GMP/GDP Inspection:** Assessment of the compliance of a site with the applicable GMP/GDP regulations and filing commitments performed by officials of a competent (local/international) health authority.

**Hybrid Inspection:** An inspection that is conducted remotely in part and also includes an in-person presence and on-site visit by the inspecting body. The remote and on-site aspects may be concurrent or at different times (e.g., a portion of the inspection is performed virtually or desktop, followed by a second portion on-site). The combination of remote and on-site components may be captured in a single inspection report or in separate reports.

**Remote Inspection:** An inspection in which the inspector is not physically present at the site(s) where the activities subject to the assessment take or have taken place. Remote inspections may take several forms, including desktop, virtual, and hybrid.

**Video Communication Platform:** Web-conferencing software that allows participants to conduct or attend meetings via the internet. This is also known as online meeting software or videoconferencing. The video communication platform enables remote meetings based on voice over internet protocol (VoIP), online video, instant messaging, file-sharing, and screen-sharing.

2 Some regulatory authorities may use documentation requests and review in advance of an inspection or to update a company's risk profile.

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**Virtual Inspection:** In addition to sharing documentation as in any other remote inspection, virtual inspections utilize technologies such as live or real-time streaming video, screen-sharing, or other means of real-time communication. Virtual inspections may include virtual tours of aspects of the inspected site. These inspections also may be referred to as distant assessments or remote interactive evaluations.

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# I. Planning and Notification of a Remote Inspection

## Topic A.1: Types of Inspections Performed Remotely

### Problem Statement

What factors might health authorities consider in determining whether a remote inspection of a site is appropriate?

### Recommendation

Health authorities can use risk assessments to determine the suitable use of remote inspections, including desktop and virtual inspections, on a case-by-case basis. Authorities will determine the level of risk based on several factors including results of previous inspections, availability of historical information and inspection reports by other trusted inspectorates, complexity of the manufacturing processes, and criticality of the products being manufactured. Desktop inspections often are more suitable for less critical assessments, while on-site inspections may be necessary for more critical inspections. External factors also may be considered, and an inspectorate may decide to use remote inspection tools for products or sites related to a public health need when an on-site inspection is not possible (e.g., during a pandemic or due to prioritization of regulatory resources).

In determining the appropriate format and tools for an inspection, the inspectorate also should consider any technological limitations and the processes, activities, or sites that can be assessed remotely with the technological tools available. For instance, inspectors may not be able to see the "full scene" virtually due to technological, security, and/or safety constraints. Wi-Fi network coverage may be limited in certain parts of the inspected facility, and some communication devices cannot be used in areas that present a risk of explosion. At the same time, sites may find alternative solutions to certain challenges. While it may be impractical for sites to convert certain documents, such as logbooks or bench books, to a format suitable for electronic sharing, for example, sites can display such documents during videoconferences using a camera specifically designed for the purpose, as discussed in more detail in **Topic I.D.1 Hardware Requirements**.

Health authorities may determine that some inspections cannot be conducted remotely. For instance, some health authorities may determine that for-cause inspections, some follow-up inspections (e.g., following a warning letter or equivalent outcome), or some preapproval inspections (PAIs) might not be suitable candidates for remote inspection at this time. Future technological developments and increasing experience with remote inspections might change these assessments over time.

If some inspection objectives can be achieved remotely, a hybrid inspection may be used. In this scenario, the remote portion of the inspection, whether desktop or virtual, can partially address the inspection scope and a later on-site inspection can be used to complete the assessment.

The legal and regulatory framework in some countries may not allow remote inspections.

### Rationale

While, theoretically, each type of inspection can be performed remotely, the level of scrutiny possible in a remote inspection may not always be equivalent to an on-site inspection. Remote inspections are heavily dependent on the technology available (e.g., use of live stream, high-resolution videos) and scope of the inspection (e.g., physical manufacturing facility versus quality management system (QMS) documentation review).

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## Topic B.1: Notification

### Problem Statement

How can an inspectorate provide notification of a remote inspection? Should this be provided in advance?

### Recommendation

For a remote inspection, the health authority may provide notification through emails, phone calls, or periodic teleconferences and meetings.

Because both parties must prepare and test the document-sharing platform and, in the case of virtual inspections, the video communication platform, the inspector is strongly recommended to provide adequate advance notification of each remote inspection. Advance notification also is necessary to allow the inspected site to prepare documentation as well as on-site and remote resources and logistics. This preparation time will allow for a smoother and more effective inspection and will help prevent frustration and lost time over technical issues during the inspection. For virtual inspections, in particular, the inspector should notify the site of the anticipated starting date as early as possible.

For a virtual inspection, additional pre-inspection requests may be sent to the inspected site, in addition to or embedded in the usual documentation request, to be completed within a specified timeframe (e.g., within two calendar weeks of receipt of the request). This additional information will assist in determining if the site is ready for a virtual inspection.

For desktop inspections, the request for documentation signals the beginning of the inspection, whereas the beginning of a virtual inspection is denoted by the opening meeting.

### Rationale

A key difference between a remote inspection (whether virtual or desktop) and an on-site inspection is the sharing of data and documentation with inspectors using electronic methods. In an on-site inspection, binders of paper documents can be easily provided to the inspectors for their review, and tours of the facility do not require the preparation and sharing of videos. To facilitate electronic sharing of information in a remote inspection, however, appropriate communication channels must be established and tested. Data and documentation may need to be prepared in advance for electronic exchange. For example, paper documents may need to be scanned, documents may need to be translated into the language of the inspectorate if requested, and data may need to be extracted from computerized databases. In addition, the facility may need to prepare for live tours of requested areas and prerecorded tours of areas where live tours are not possible, including areas without a Wi-Fi signal. Thus, advance notification to the inspected site is important in achieving a satisfactory remote inspection experience.

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## Topic B.2: Verification of an Inspector's Identity

### Problem Statement

In the remote setting, how can a site verify the identity and affiliation of a person presenting him/herself as a regulatory inspector?

### Recommendation

Verifying the identity and affiliation of any inspector conducting a remote inspection is critical. The remote context makes this verification more difficult than it is during an on-site visit, where the inspected site personnel can view the visitor's official credentials in person.

The inspected site should ensure that the regulatory inspector or other representative is legitimately employed by the health authority he/she represents. The site should prepare in advance so that it can conduct this verification expeditiously without unreasonably delaying the inspection. Companies can take the following steps:

1. Become familiar with published policies and procedures followed by relevant health authorities. Check websites or inquire directly about the normal process and sequence of events that the health authority uses to initiate each type of remote inspection. Determine whether the health authority has defined a protocol for inspectors to validate their affiliation. Many health authorities provide official documents to verify their employees' affiliation. Some require their inspectors to display official credentials to the site's owner, operator, or agent-in-charge.

2. If personnel at the inspected site have any reason to doubt the authenticity of the source of an inspection notification, contact the health authority by email or telephone to confirm it.

3. Report any suspected impersonation of a health authority official directly to the health authority's management, providing as much detail as possible so that the health authority can take action in the unlikely event of impersonation.

### Rationale

While reported instances of impersonation of regulatory inspectors are rare, they have occurred. For example, in December 2009, the U.S. FDA issued a press release warning of an impersonation scam ***(1)***. By impersonating an inspector, a person could engage in industrial espionage and gain access to records, trade secrets, and confidential information.

The company can protect itself by verifying the legitimacy of the inspector. While it may be more challenging to confirm the inspector's legitimacy in the remote context, steps should be taken to verify identity and authenticate affiliation before releasing records, answering questions, or enabling views of premises, equipment, and processes.

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## Topic C.1: Remote Inspection Team Formation

### Problem Statement

What team should the inspected site assemble to prepare for a remote inspection? What is the role of the Remote Inspection Host?

### Recommendation

The first point of contact for the health authority usually will be the same person that would be notified of an on-site inspection. The first point of contact should promptly involve the personnel responsible for establishing a suitable communication channel for the exchange of confidential information. The communication channels for information-sharing should be agreed by the inspected site and the inspectorate. Early communications may be made primarily through telephone and email, but document-sharing may require additional structure and protections, as discussed in more detail in **Topic I.D.2 Planning to Share Information**.

To ease the communication process, the inspected site should identify one person to serve as the "Remote Inspection Host"/Single Point of Contact. The inspectors should use the Remote Inspection Host as the single point of contact for all inspection-related communication before, during, and after the inspection. This person must have or be supported by adequate authority to ensure that the remote inspection can proceed smoothly. For instance, the Remote Inspection Host must be able to quickly obtain support from the Information Technology (IT) or similar department to solve any problems that arise during the inspection. The Remote Inspection Host described in this paragraph is not necessarily, and often will not be, the Responsible Person in a GDP setting or the Qualified Person in a GMP setting.

If several inspectors will be involved, it may be useful to have multiple Remote Inspection Hosts. The inspection team should indicate the number of inspectors that will be involved or the number of workstreams that will be necessary in order to allow the inspected site to prepare both personnel and technology. See **Topic II.B.1 Managing Multiple Workstreams** below for further discussion of this point.

The team that manages the remote inspection for the inspected site should be the same as for an on-site inspection. The team should include the inspection management personnel, the site leadership team (at least for opening and wrap up/closing meetings), subject matter experts (SMEs), local operating company representatives as applicable, IT support, and representatives of global functions when required. Members of the inspection management team may be able to participate from off-site locations if they are not also on the core operational team described below. The inspection management team also should consider designating back-up personnel in case named personnel are unavailable due to connectivity issues.

During the remote inspection, a core operational team, including a designated leader, Remote Inspection Host, and IT support, should be present at the inspected site. The designated leader and the Remote Inspection Host may be two different people, with the designated leader playing a leadership and strategic role and the Remote Inspection Host facilitating the communication process. The two must work together.

The on-site core operational team will be responsible for document and data management during the inspection. In a virtual inspection, this team also will manage video requests and virtual facility tours. This team should develop and optimize an efficient process to locate and scan, arrange for translation as necessary, put into electronic format, review for completeness and accuracy, and upload onto the document-sharing platform any document that may be requested during the inspection. This process is important to ensure a smooth workflow.

The core operational team must fully respect required controls for health and safety, e.g., during a pandemic virus outbreak. When health and safety rules limit the number of people that may be present at

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a site or limit access to the manufacturing areas, the inspected site should mention such constraints up front and ensure that inspectors are aware that these constraints may lead to slower responses.

Certain personnel may be able to participate from off-site locations. If the QMS documentation and batch manufacturing and control records are available electronically, for instance, personnel may not need to be physically present at the inspected site to present this data. Similarly, support on specific topics can be provided by company experts located in different geographic areas, regardless of the location of the records and documents.

The inspected site should provide the inspector a list of anticipated participants in the different steps of the inspection. The list should include each participant's job title as well as their role in the remote inspection process. Similarly, the inspector should communicate the presence or participation of all inspection team members involved in a virtual inspection. A complete list of all personnel that participated should be provided by both parties at the conclusion of the inspection.

## Rationale

The remote inspection format inherently allows more people to participate from diverse locations on behalf of both the health authority and the inspected site. With this increase in the number of participants compared with an on-site inspection, personnel roles and responsibilities should be clarified at the beginning of the process. When health and safety conditions require (e.g., during a pandemic), the number of people present on site should be the minimum necessary to handle requests for data and information that is available only on site.

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## Topic C.2: Planning the Agenda

### Problem Statement

How might an inspection agenda be different in the desktop or virtual setting?

### Recommendation

The review process in desktop and virtual inspections in some cases may be less efficient and more challenging. Virtual inspections generally range in duration from a few days to 2 weeks, with intense support required during this period. The overall time span to complete a virtual inspection can be double or triple that of an on-site inspection if the amount of time spent in a virtual inspection on any day is less (e.g., 4 hours instead of 7), however. In contrast, the timeframe for desktop inspections tends to be especially long, as they are iterative in nature, but the intensity of engagement with the inspector may vary during the process. In desktop inspections, inspectors generally will request further information, data, and documentation as they review initial materials, and the inspected site will subsequently respond to these additional requests. Desktop inspections can take more than a month to complete.

As in an on-site inspection, the agenda is a critical aspect of an efficient virtual inspection. A clear agenda allows the site to prepare documentation and arrange interviews and virtual tours to minimize waiting time. Both parties should recognize that departures from the agenda may be necessary to complete coverage of areas, to follow leads or areas of interest, to observe operations in real time when they occur, and to accommodate availability of key personnel. If the inspector does not provide a comprehensive agenda in advance, the two parties should discuss potential agenda complications such as operations schedules and times that key staff may be unavailable.

The agenda for a virtual inspection may be different from an on-site inspection agenda. An agenda that alternates short live conversations with periods of off-line document preparation and review can help prevent fatigue and inattention. Periods of informal discussions with inspectors are valuable to ensure the inspector's satisfaction with the information provided and with real-time responses to any issues raised. Hourly short breaks allowing participants to move around and/or turn away from the computer screen also are important to maintain well-being.

Time zones also must be considered when inspectors and inspected facilities are in different areas of the world. Video communications and interviews should be conducted during the normal working hours of both the inspected facility and the inspectors, to the extent possible. If this cannot be done, the inspectors should consider shifting their working hours to those of the inspected site, as this helps ensure that the inspection progresses quickly, limits data integrity risks, and allows the inspectors to build rapport with site personnel. That said, both parties should respect that working at unusual hours can cause significant fatigue for participants.

It is helpful for the inspector to share the date on which the virtual or hybrid inspection will start and end, if possible. If no specific dates can be shared, a general expected timeframe would be useful. Because health authorities take different approaches depending on the type of inspection, it can be difficult for inspected sites to make predictions about this timing. In addition to clarifying the expectations for both parties, this information allows the inspected site to ensure that necessary personnel will be present.

A daily wrap-up may be built into the agenda to allow the parties to discuss any technical issues or necessary adjustments to the agenda, as well as to ensure clear understanding of the scope and topics to be presented and discussed the following day. These wrap-up meetings are discussed in more detail in **Topic II.E.1 Use of Daily or Intermediate Wrap-ups**.

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## Rationale

The review process in a virtual inspection is inherently different than in an on-site inspection in several ways:

- The review process might last longer than it would in an on-site inspection. For instance, fewer inspection hours per day, or delays while working with a new platform or tool, may require the virtual inspection to be scheduled for a longer time period.
- Additional time may be required for the clarification of some topics due to the different format for the interactions between the inspector and site personnel.
- The inspected site may need more time to provide supporting documentation or to fulfill additional requests.
- Less time is available for informal "hallway" conversation, making it more difficult to build rapport and relationships.
- Communication style and cultural differences are more difficult to understand remotely, and the lack of body language is a further complication to remote communication. Special attention should be given to cultural differences that may affect the communication between the personnel at the inspected site and the inspectors.
- If translated documents are shared in advance of the virtual inspection, follow-up emails may be necessary to clarify the content for comprehension. Clarification of translated language can be more efficient in a live conversation than through email.
- The need to closely respect the planned agenda, which may be less flexible in the remote context, can create stress.
- Being seated and using the screen continuously can be challenging and tiresome. Differences in time zones may require work outside of regular working hours, which affects well-being.
- Potential connection issues may impact the quality of audio and video communications while also possibly delaying the inspection.

In a virtual inspection, it is important to develop a realistic agenda that considers these challenges and constraints. To avoid unnecessary delays, the sequence of the topics to be assessed must be clear, with a waiting list of subjects to be discussed.

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## Topic D.1: Hardware Requirements

### Problem Statement

What hardware is required or recommended for a virtual inspection?

### Recommendation

The basic requirements for a virtual inspection follow:

- **Computer:** All participants must have a computer with a reliable internet connection of a speed that matches the requirements of the technological tools and communications platforms to be used.
- **Webcam or conference camera:** A videoconferencing camera or a camera built-in to the computer can be used.
- **Audio:** Computer audio must be suitable. The speakers and microphone for each participant should be of appropriate quality.
- **Intranet or virtual private network (VPN) connectivity at inspected site:** The inspected site should test connectivity from all locations involved in the inspection, including those of key staff participating remotely, in advance of the start date to ensure that all software applications will be accessible.

The following items are suggested:

- **Telephone line:** A separate telephone line should be designated as a backup means of communication in case of an internet or computer software outage. A cellular or landline phone may be appropriate as long as both parties clearly understand when it is to be used. The telephone line must be accessible at any time, have good reception in the areas in which it would be used, and include a speakerphone option.

- **Larger monitor:** A 20-inch or larger monitor is highly recommended to reduce eye strain, assist in quicker review of the documents, and possibly allow for review of multiple pages or documents at a time. Participation in an inspection by tablet or handheld device is strongly discouraged due to the size of the screen.

- **Multiple monitors:** Inspected site personnel and inspectors may find it helpful to use multiple computer monitors. This allows them to manage documents on one monitor while participating in video communication on the other, and to increase the font size of or zoom in to documents for easy readability.

- **Document camera:** A document camera is a real-time image capture device that functions as a high-resolution web camera, mounted on an arm to facilitate the display of a document or object with magnification if necessary. Document cameras can be used to show real-time images of such documents as in-process records, logbooks, or test data. These cameras can enhance the inspection and reduce the burden of additional copying and transmission of records that may not be needed by the inspectorate.

- **High-speed scanner:** A high-speed scanner can be helpful in all types of remote inspections as the inspected site prepares documents for sharing.

If a virtual tour will be provided as part of the virtual inspection, the following points should be considered:

- Several options are available for providing virtual tours.[3] Virtual view tools such as 360° cameras or virtual reality goggles may be used. Other companies may use a laptop or mobile device. The device must be appropriate for the area being shown (e.g., manufacturing area or laboratory) and to provide the requested information.

3 CCTV feeds might be used in specific circumstances, if requested by inspectors, appropriate to show GMP areas, and permitted under national law and company procedures relating to security and data privacy.

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- Care should be taken to ensure that the camera produces a stable picture that is easy to view. For this purpose, the camera can be mounted on a movable trolley or gimbal. The site should be prepared to remove the camera from any mounting to allow show equipment and facilities from all angles requested by the inspector. Unstable, wobbly pictures that occur when a camera is hand- or body-mounted, or when a trolley rolls over uneven floors or stairs, are difficult for inspectors to view and can induce nausea and, in some viewers, vertigo.

- Additional audio tools may be necessary for the person operating the camera during the virtual tour. The camera operator needs to be able to hear the inspector's directions on where to move or point the camera, even in areas with significant background noise. The camera operator also must be able to hear and respond to questions from the inspector during the virtual tour.

- In areas of the facility where the Wi-Fi signal may not be strong enough to broadcast a live virtual tour, consider whether it might be useful to install devices to amplify the signal.

All devices used to support an inspection, including laptops and mobile devices, should be approved for use within the area of intended use, including classified cleanrooms and hazardous areas, from a quality and safety perspective. Controlled/explosive zones, such as manufacturing areas involving flammable solvents or explosive gases, may require special precautions.

## Rationale

The hardware used in an inspection can significantly contribute to the efficiency and effectiveness of the inspection. Hardware that is well-suited to the purpose also can reduce strain on all participants in the inspection process. The utility of a virtual facility tour is dependent upon the camera operator's responsiveness in providing a stable image of requested aspects of the manufacturing site.

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## Topic D.2: Planning to Share Information

### Problem Statement

What is the most suitable process for sharing documentation in a remote inspection? In a virtual inspection, what is the most suitable process for conducting interviews and providing virtual tours of the facility?

### Recommendation

In preparing for a remote inspection, the parties should consider confidentiality, security, and data protection as well as the technological tools available. All communication channels must be agreed by both parties and provide the appropriate level of protection for the exchange of confidential and sensitive information. Both parties should involve their IT department (or equivalent) in these discussions.

Regardless of the technological tools selected, it is essential that all participants test and become familiar with the tools in advance of the inspection.

#### *Documentation*

The inspected site should prepare documents in digital format. Documents shared electronically should be:

- Formatted to be searchable, to the extent possible. Searchable documents are essential to an efficient inspection because inspectors often may be looking for certain words, phrases, or information within a document.
- Provided in a clearly organized manner, consistent with the document request.
- Identified in a way that enables the inspector to understand each document's contents from its title or file name without having to open the document.
- Provided in a digital form that can be opened with widely-used software applications. For instance, documents in.pdf or.csv format can be opened using many different types of software.

The options for sharing documentation range from secured email to secured cloud storage solutions restricted to specific participants and made available only during the inspections. Existing gateways, such as those used for the submission of regulatory data, also can be used as applicable (e.g., for PAIs, when the inspection is part of the approval of a registration dossier).

In general, the use of a document-sharing platform is recommended. Materials shared in this way may include scanned copies of documents, word processor documents, spreadsheets, machine/software generated documents, and videos. The document-sharing platform should provide the appropriate level of security while also allowing auditors to easily locate, review, and annotate documents. The inspector and the inspected site should discuss the inspector's ability to download or print the documents from the document-sharing platform.

If mutually agreed, documents also can be shared in real time during the inspection through the video communication platform. For instance, a document camera, which is a video camera on a tripod specifically designed to display documentation live, can be used to present paper documents involving many pages, such as validation documentation. A best practice is to use a document camera for sharing discrete, specific information from a hard copy document (e.g., one page from a logbook).

Documents may be shared by email using an email address specified for inspection communications. This should be avoided generally, as it is difficult to manage the large number of emails that would be necessary to provide the required information. File size may be limited if sharing via email.

Documents also might be shipped through traditional mail or courier services, either in paper form or on electronic storage media. Express, traceable shipping methods should be used. However, this

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usually should be a last alternative, as it might require more time and paper documents are not easily searchable.

The inspected site should ensure that any electronic files are accessible by the inspector. The inspected site should ask the inspectors to acknowledge that they have received the provided documentation and are able to access it, if the document-sharing platform does not automatically record this information.

As in an on-site inspection, documents that are provided electronically should also be kept on site in a well-organized manner in case it is necessary to cross-check or provide the document to the inspector in a different format during the inspection.

#### *Remote Interviews and Virtual Tours*

Livestream videos and videoconferences are the most effective method of conducting remote interviews and virtual tours of facilities. Video communication platforms can be used for many aspects of a virtual inspection, including video- and audio-conferencing and chat. Examples are GoToMeeting™, Microsoft Teams™, Cisco WebEx™, and Zoom™. As the party sharing information, the inspected site may recommend a platform. Health authorities, however, may limit the platforms that inspectors are permitted to access.

The video communication platform should allow for appropriate number of separate workstreams, based on the number of inspectors that will participate. In general, the platform should allow for at least three simultaneous workstreams. The selected platform also should allow participants to join by telephone if necessary. Alternative technologies also should be defined in case of unexpected system failure.

The inspected site should consider preparing a video giving a general overview of the facility, which could be made available during the opening meeting if requested by the inspector. If the inspector would like a tour, the inspected site can provide one in real time with an operator moving around in the facility with smart glasses or camera, as discussed above in **Topic I.D.1 Hardware Requirements**. Alternatively, still photos or pre-recorded video could be used, particularly for areas that cannot be toured live.

Note that some materials developed by the inspected site for the inspection might be considered GMP records, which would be subject to additional protections. The purpose of the record defines whether or not it is a GMP record. A GMP record is a record that contains information related to an activity subject to GMP regulations and required by the GMP regulations to be retained as historical evidence of the activity and/or to support quality, compliance, safety, and efficacy of the product. Therefore, most videos, photos, presentations, and interview records (if recorded) that are created specifically to support a remote inspection would not, in most cases, be GMP records. However, inspected sites should review the status of the documents with legal counsel in advance, develop a business process for handling them, and determine minimum retention time.

## Rationale

Modern technological solutions allow for real-time interactions that closely mimic those of an on-site inspection, and technology will continue to advance. However, health authorities and manufacturers may have different priorities and preferences in selecting communication tools.

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## Topic D.3: Planning for Effective Use of Technology

### Problem Statement

What additional considerations might improve the use of technology in a virtual inspection?

### Recommendation

In selecting technology for a virtual inspection, two points are important to keep in mind:

- **Technology Overkill:** Technology is intended to help the successful execution of the inspection, not to showcase different options. Companies being inspected may range from large, well-resourced global companies to smaller organizations in remote locations with less-developed infrastructure. Balance should be maintained in technology options.

- **Technology Exploration:** Exploring new technologies or options during the inspection is not recommended, as these options may require installation and training time or may unnecessarily delay the inspection.

Because many factors can impact the actual performance of the technological tools during a virtual inspection, all technology must be tested in advance, including the document-sharing platform, video communication platform, and Wi-Fi network (for robustness). The connectivity of the devices to be used also should be tested in all areas of the site and remote locations. Each participant should have a reliable internet connection of a speed that matches the requirements of the video communication platform, to allow for consistent and stable audio and video. A half-hour real-time test several days before the inspection can be helpful to check for and resolve any issues, including any limitations on software use based on geographical region, elapsed time, and number of users.

In discussing the software and applications to be used, the two parties should be sure to identify any known or potential compatibility issues, such as antivirus software, that is known to inhibit the use of certain applications. The inspected site may recommend an internet browser to be used during the inspection, as the site may have the most knowledge of any software compatibility issues relating to specific browsers.

If site personnel will be demonstrating site-specific software during the inspection, they should verify in advance that the software is accessible from the computer system that the personnel will use during the inspection. Software specific to the manufacturing location or laboratory, including software used for issuance of batch records, temperature and pressure monitoring, and other integrated manufacturing software systems, may not be accessible from all company computers or may reside on networks that are decoupled from the internet. Advance preparation will ensure that personnel can participate in the inspection from the manufacturing location or laboratory, or that they can take other necessary steps.

Whether inspectors will be given direct access to data systems also should be discussed early. As covered more fully in **Topic II.C.1 Inspector Access to Systems**, allowing inspectors direct access to GxP systems is not recommended due to the need for training and the risk of creating errors within the systems. Instead, a site expert can log into the system, share his/her screen through the video communication platform so the inspector has a clear view of the data and information, and allow the inspector to direct the expert's movements through the system.

### Rationale

While it is important to select effective technological tools for a virtual inspection, it is equally important that site personnel and inspectors are able to use the tools, and that the parties have consistent expectations about how the technologies will be used.

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## Topic D.4: Pre-Inspection Agreements

### Problem Statement

What agreement should the parties reach before the inspection begins?

### Recommendation

The parties should agree upon, and describe in writing, the equipment and technology to be used in the remote inspection, including the document-sharing platform and video communication platform described in **Topic I.D.2 Planning to Share Information**. An agreement should be reached as far in advance of the start of the inspection as possible.

The parties also should define clear rules for the sharing, retention, and disposal of any electronic records as part of the inspection process to ensure appropriate handling of confidential and/or sensitive information and compliance with applicable laws. The inspector should ensure that the health authority will protect from unintended use the documents shared by the site, and confirm this with the inspected site. The inspected site also may request that the health authority identify all individuals who will access the information provided by the inspected site, including all participants in inspection-related review activities.

The use of technology in a remote inspection is limited by the need to properly protect data and information. Cybersecurity issues should be considered carefully. Both parties should:

- Confirm the applicability of confidentiality controls to prevent disclosure of confidential commercial, personal privacy, or trade secret information
- Verify the security of the communication channels

Legal counsel should be consulted during the planning, as local law may require additional agreements from both sides and, possibly, from the corporate office of the inspected entity before initiating video communication and video tours. For instance, local law may prevent the recording of sound and images or may require authorization to capture or use images of individuals. In some situations, security requirements may not allow the use of video. Where applicable, the Data Protection Officer of both organizations should assess the proposed procedures and identify, in advance, any confidential or sensitive information and its proper handling.

Legal counsel also may need to be involved in any discussions about whether the site will allow any recording of conversations, video communication, images, and screenshots and, if so, specify any limitations on recordings. The site should not record any portion of a remote inspection (whether by video, audio, or other recording) unless the inspected site has made a clear, formal request to do so, and the health authority has expressly granted permission. If the proposed video communication platform automatically records all sessions, all parties should be made aware of this fact during planning.

### Rationale

Prior agreement to the technology to be used and the protections to be given will build confidence and help assure a successful inspection by allowing the process to be more efficient. Establishing the agreement in advance will allow both parties to install and test any necessary technology before the beginning of the inspection, which will prevent delays during the inspection itself. Legal counsel for both parties should review the agreed-upon framework.

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## Topic E.1: Preparing Necessary Documents

### Problem Statement

What information and documents should the inspected site prepare in advance of a remote inspection?

### Recommendation

Health authorities expect sites to be inspection-ready. As health authorities increasingly use remote tools for inspections, sites should maintain appropriate documentation to support a remote inspection on short notice. The inspected site should prepare and organize all key documentation before the inspection begins.

Many sites have developed a standard list of documents that are likely to be requested in an on-site inspection. This list is equally applicable in remote inspections. The documents can include site organization, size, products manufactured, list of standard operating procedures (SOP), deviation reports, site master files, biological product deviation reports or field alert reports and out of specification results, corrective and preventive actions (CAPA), applicable Annual Product/Quality Reviews that include details on change controls, complaints, recalls, batches manufactured, aseptic process simulations with outcomes, environmental monitoring data, trends for utilities (e.g., water, steam, gas), and sterility test failures.

In advance of a remote inspection, the inspected site should also prepare the documentation that an inspector would normally review on-site or may request at a later point. Typically, this additional documentation might include:

- Data analysis (e.g., deviation recurrence analysis, equipment failure rate, CAPA effectiveness review) and/or lists that can be searched or filtered
- Comprehensive inspection history, especially for the first remote inspection of the site and for-cause inspections
- History of or changes to key SOPs since the last inspection, along with training records
- Information related to relevant investigations.

**Appendix 1** provides a list of documentation covering most of the areas and subjects that typically may be assessed in an inspection. The list may vary depending on the purpose of the inspection, e.g., PAI versus GMP surveillance.

If a virtual inspection of the manufacturing operations, facilities, and equipment will include video or photos of the site, it can be helpful to orient the inspectors by supplying site schematics or layouts, drawings, and/or process flow diagrams. The location from which any videos or photos were taken should be marked on the site schematics. Using the drawings already submitted to the agency, which are included in the Site Master File, should be considered. These documents may be readily available to the inspector and their use might avoid data integrity concerns.

By timely communicating, in detail, the documentation that the inspectorate wishes to assess as part of the inspection, the inspector can increase the efficiency and effectiveness of the remote inspection. This early communication can minimize the possibility of delays mid-inspection as the inspected site locates an additional document or data, translates it as needed, and prepares it for sharing. Providing a list in advance also reduces the risk of data integrity issues that may arise from mid-inspection preparation of documents for sharing.

### Rationale

Early preparation is especially important for remote inspections, in which rapid or real-time translation and review of documentation may not be feasible or convenient. While the initial preparation may involve significant effort, the result should be a smoother and more efficient inspection.

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## Topic E.2: Sharing of Inspection Reports

### Problem Statement

Will a health authority share its inspection reports with other health authorities? Can an inspected site share inspection reports with inspectors from other authorities to facilitate the inspection?

### Recommendation

Inspectors may leverage information and data that other health authorities have collected through on-site inspections. This information can be very useful in supplementing the information obtained during a remote inspection. Inspected sites should anticipate that inspectors have reviewed other health authorities‘ inspection reports.

Confidentiality arrangements in place within the Pharmaceutical Inspection Co-operation Scheme (PIC/S), or between mutual recognition partners and other collaborating health authorities, allow health authorities to share and access GxP information on international organizations and supply chains involved in the development, manufacture, and supply of medicines. The PIC/S inspection reliance procedure is particularly important, as it sets out a framework for participating authorities to rely on the inspection information provided by the network ***(2)***.

Inspectors also may ask the inspected site to share inspection reports and GMP certificates from other health authorities. For these documents, the same rules apply as in on-site inspections. Some inspectors will expect the reports to be shared. Inspectors also may request evidence of appropriate post-inspection CAPA implementation. Highlighting the site's implementation of any agreed-upon post-inspection actions can help demonstrate responsiveness and agility.

Some health authorities, however, may require that they be consulted or informed before their inspection reports are shared. The site also should verify whether the relevant authorities have mutual recognition agreements (MRA) or if the document is subject to freedom of information legislation. If no MRA is in place and the document is not subject to freedom of information, or if the report includes a nondisclosure clause, personnel at the inspected site should consult with legal counsel and ask the issuing authority for approval to share the information.

### Rationale

Today, there is a significant convergence in worldwide regulations and standardization in the execution of inspections by global inspectorates. Many inspectors already use information provided by other regulators and rely on the outcome of inspections(s) made by other health authorities, as mentioned in section 2.2 of the PIC/S guidance:

> Informed decisions on the GMP compliance of a manufacturing facility can be made, in certain circumstances, based on the outcome of work by another regulatory authority or authorities. Consequently, it is possible (outside the established framework of Mutual Recognition Agreements, or equivalent and where legal requirements allow) for Inspectorates to identify specific instances where an onsite inspection of a manufacturing facility in an overseas territory is not required because an acceptable level of GMP compliance has been confirmed and assured by another regulatory authority or authorities. ***(2)***

In this way, inspectors can supplement their own remote inspection with data collected in the field by other qualified inspectors.

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## Topic E.3: Timelines for Sharing Documentation

### Problem Statement

What are the timelines to respond to the pre-inspection notification and documentation request? Is it possible to send the requested documentation in multiple shipments?

### Recommendation

Most authorities require inspected sites to provide documents for pre-inspection (in the virtual or hybrid context) or inspection (in the desktop context) within ten working days or two calendar weeks. In general, timelines for responding to the first notification and to additional requests should be negotiated with the inspector, considering the quantity of information and data requested, their format, and any translation requirements. The timeline negotiated with the inspector also will depend upon the size of the site, the complexity of the operations, the inspection scope, and the level of risk associated with the inspection.

It may be necessary or helpful to share the information and documentation with the inspectorate in several different bundles to avoid delaying the review process. To prevent misunderstanding, the inspected site should clearly inform the inspector of its intention to send several different packages of information.

If the inspector numbers its requests, the inspected site should provide documents in folders that use the same numbering system to facilitate review and confirmation that all documents have been provided. If the inspector does not use a numbering system or another clearly repeatable method of organization, the inspected site should bundle the documents by topic to facilitate review.

The site also might consider including notes with documents as necessary to help the inspectorate understand the context.

In the rare circumstance that documents or data are not available, the inspected site should communicate this fact to the inspectorate in a timely manner with explanation.

### Rationale

The quality and clarity of the documentation provided is critical to an efficient and effective inspection. Therefore, the inspector should allow the inspected site sufficient time to prepare. Usually, two calendar weeks may be sufficient for the inspected site to provide the first set of documentation requested.

While as much of the documentation as possible should be provided in advance of a remote inspection, some documentation will be requested and shared during the course of the inspection, and may be requested with a short turnaround time. These additional requests are discussed in **Topic II.D.1 Responding to Requests**.

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# II. Remote Inspection Realization

## Topic A.1: Use of Real-Time Interactions

### Problem Statement

What is the most efficient and effective way to use real-time interactions in a virtual inspection? Is it possible and recommended to include multiple people or subject matter experts (SME) in the live sessions?

### Recommendation

As discussed in **Topics I.D.2 Planning to Share Information** and **I.D.4 Pre-Inspection Agreements**, for a virtual inspection, the inspector and inspected site should agree on a video communication platform. Prior to the inspection, all involved personnel should become familiar with the IT tools to be used.

Good meeting management is essential. It is preferable to establish one meeting link or connection for the entire day, which all attendees may use, rather than creating a separate meeting link for each subtopic to be addressed during the day. Inspectors and inspected site personnel can easily leave and rejoin the ongoing discussions, although attendees should always be introduced upon joining. If attendance is not carefully managed, however, the inspected site could be overloaded with questions and requests, including potential duplication from those who participate only intermittently.

For each workstream, one or two members of the inspection team should ask all questions, and one person at the inspected site should coordinate the discussion as a moderator. Also refer to **Topic II.B.1 Managing Multiple Workstreams**.

Several people (such as SMEs) from the inspected company could join the live sessions to be ready to support the ongoing discussion. The inspected site should control the flow of people joining the live session by using a virtual waiting room, if possible. A person who is not directly interacting with the inspectors should be assigned to monitor the virtual waiting room.

The video communication platform can be used for written conversation and for sharing documents and data. A live chat channel with the inspectors (e.g., via Microsoft Teams® or Slack®) also may be used to facilitate communications if needed. It may be helpful, for instance, for the inspector to drop notes about requests, such as deviation numbers copied from provided spreadsheets, into this chat channel to ensure that the correct information is available to all. This channel should be agreed upon by the parties in advance.

The inspected site and inspectors should not communicate outside of the agreed-upon communication platform(s). Additional informal communications via text or mobile phone application, for instance, should not occur, as these would be difficult to track and document and may cause confusion.

As in on-site inspections, an internal chat-link for site personnel may be useful to facilitate internal information-sharing as well as to verify the timing of topics to be presented. The internal chat-link could be set up through an online messaging platform, text, or email.

In planning for virtual inspections, training personnel on virtual meeting discipline may be helpful. This training could include:

- **On-camera discipline:** Training should address personal attire, background screen selection, and prevention of accidental entry of family members and colleagues during video communications. Personnel should keep their cameras on while in the virtual meeting room, notify other attendees when turning off the camera, face the camera properly, and understand expectations around eating and drinking on camera. Discuss and decide whether cameras may be turned off during document review periods.

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- **Microphone discipline:** Microphones should be muted when not in use and background noise avoided. Particular attention should be paid to feedback loops that result in unacceptable echoes. Often, this may result when a participant logs in from more than one device. If muting another person's microphone is necessary, inform the person before doing so.

The inspected site should log all questions asked by the inspectors, including questions asked through live conversation and in writing, as it would during an on-site inspection. The log allows for visibility, sharing, and tracking of the questions within the inspected site.

## Rationale

To prevent disturbances and interruptions, virtual inspections require enhanced coordination as well as discipline in communication.

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## Topic B.1: Managing Multiple Workstreams

### Problem Statement

What tools can an inspected site use to manage multiple workstreams in a virtual inspection?

### Recommendation

Multiple workstreams may be required, for example, when different inspectors are reviewing different material in a single inspection. In this case, the inspected site should create a separate meeting link or virtual breakout room for each inspector or workstream and should assign a remote inspection host for each inspector that needs an independent workstream.

The inspected site may use one full-day meeting link for full-team conversations including opening and wrap-up discussions, one full-day meeting link for Inspector A, and one full-day meeting link for Inspector B, with each to be used only as necessary. Alternatively, some video communication platforms offer virtual breakout rooms, which can be used in the same way. As with all other technologies, inspected site personnel should become fully familiar with these tools in advance of the start of the inspection.

When multiple workstreams exist, the remote inspection host must coordinate to ensure that information is shared appropriately and to avoid duplication of requests. The inspected site should create a process that allows all requests for documents and other information to be visible to all members of the inspection team. The secured document-sharing platform might be used for this purpose. When dealing with multiple workstreams, all questions asked by inspectors must be logged, as discussed in **Topic II.A.1 Use of Real-Time Interactions**, and the question log must be accessible to all remote inspection hosts.

### Rationale

Due to the higher complexity of the virtual inspection process, careful attention to and oversight of the different workstreams is essential. With time and experience, inspected site personnel and inspectors alike will become more familiar with the application of the different tools and processes, which will lead to a smoother inspection process.

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## Topic C.1: Inspector Access to Systems

### Problem Statement

Should an inspector be allowed to access GMP/GDP operational systems?

### Recommendation

If an inspector requests direct access to a GMP/GDP operational system, such as a laboratory information management system or building management system, the response should be the same as it would be during an on-site inspection. Inspectors should access systems only through trained site personnel, due to the risk that direct access by an external user could create errors in the system.

Expert site personnel should access the system, then allow the inspector to view the requested information through screen-sharing. The inspector can ask or direct the expert to obtain or display specific information and data from the system. The inspector also can direct the expert on what to do, what commands to execute, and what the inspector wants to see, provided that these activities do not create any errors in the system or create data that is not triggered by a process.

Not allowing the inspector direct access to a system will not represent an inspection refusal. Instead, it should reinforce to the inspector that proper training is needed before a person can be granted access to an electronic system.

### Rationale

An inspection should not put a GMP/GDP system at risk at any time. Access to GMP/GDP operational systems requires dedicated training and supervision and should be justified. At the same time, it is essential to operate transparently and to permit the inspector to observe the system and the data it contains within the scope of the inspection. As in on-site inspections, inspectors can be provided full visibility by directing a company expert to display specific information and data from the system; therefore, such a practice cannot be considered as an inspection refusal.

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## Topic D.1: Responding to Requests

### Problem Statement

What are acceptable timelines for responding to requests from inspectors during the inspection? What requests should be brought to the attention of legal counsel?

### Recommendation

A site's plan for responding to requests in a remote inspection should mirror the same way as working during an on-site inspection. Responses should be provided as soon as possible, consistent with the inspected site's operations. To facilitate efficient responses, SMEs and other experts in the company should be ready and available for the duration of the inspection. When it is necessary to retrieve archived documents, or when the team at the inspected site is working completely or partially remotely, the response time could be slightly longer. Communicate proactively and transparently with the inspector if some responses could take additional time to access, and agree on acceptable timing.

As discussed in **Topic I.C.1 Remote Inspection Team Formation**, the company must have sufficient personnel standing by to facilitate the handling of documents during the entire inspection, starting from the first request for documentation and records. Based on the types of requests received, personnel and resources may need to be shifted to ensure timely fulfillment of requests.

The inspected site may anticipate that legal counsel will need to review certain requests. The company should establish a strategy for these requests before the inspection begins, to help avoid delays during the inspection process. In a remote inspection, legal counsel might be consulted for:

- Requests for video or photos of personnel operating in the inspected areas, even when it might be difficult or impossible to identify the participants
- Requests by the inspector to record proceedings using photos, audio, video, or screenshots
- Any request that the inspected company believes it must decline

When sharing documents during an inspection, consideration should be given to the size of the electronic files. It may be necessary to split documentation into multiple electronic "shipments" to ensure its delivery, depending on the method used for sharing.

### Rationale

Timely responses to inspectors' requests are crucial to an efficient and effective inspection. However, challenges may arise in retrieving documentation and making documents available electronically during a remote inspection. The inspected site must communicate in a timely manner with the inspectors if the requested documentation cannot be provided by the specified due date, with justification and a target date for a complete response. Communication is especially relevant when the workforce at the inspected site is smaller than usual (e.g., during a pandemic) and when the request involves paper documents or documents requiring translation.

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## Topic D.2: Sharing Audit Reports

### Problem Statement

What should the inspected site do if asked by the inspector to share the audit reports of contract manufacturing organizations (CMOs) or suppliers?

### Recommendation

As in on-site inspections, an inspector in a remote inspection can ask for any document that will provide information necessary to assess compliance. Inspectors from some health authorities may not have legal authority to demand, or may not ordinarily review, the inspected site's audit reports of CMOs or suppliers, so it is important to understand the inspector's perspective.

In a virtual inspection, extracts of an audit report, including the overall audit outcome and the front page of the report showing execution date and scope, can be displayed to the inspector through screen-sharing or use of a document camera. Some health authorities may request that the full report be displayed. In extraordinary cases, it might be necessary to provide an electronic copy of the document.

The inspected site may need to consider whether the quality agreement with the CMO or supplier includes language about the sharing of audit reports. This aspect should be considered in advance of the inspection to avoid creating delays in communicating the information to the inspector. Moving forward, the quality agreement should stipulate that such reports may be shared and should define the timeline for notifying the CMO or supplier that the document has been shared. It is a best practice to notify the CMO or supplier before or in parallel with sharing the complete report with an inspector.

### Rationale

The request for an audit report in a remote inspection is equivalent to the request that can be made during an on-site inspection. These requests may be more prevalent in the remote setting, however, as inspectorates seek additional supporting information.

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## Topic D.3: Sample Collection

### Problem Statement

During a remote inspection, can an inspector request samples from the inspected facility?

### Recommendation

During a remote inspection, inspectors are unlikely to ask the inspected site to collect and send product samples from its facility or to collect and analyze an environmental sample (e.g., residue observed during a video tour) because inspectors normally expect to be involved in sample collection. In Europe, collection of samples is not standard.

If asked, the inspected site should clarify the conditions for collecting and sending the samples. The inspected site also should consider and mitigate any risks relating to the chain of custody. The site's inspection management team also should contact legal counsel before agreeing or declining to provide any samples.

### Rationale

Though unlikely in a remote inspection, a request for samples might be made under special circumstances. The inspected site should clarify and diligently follow the sampling and shipment procedures to ensure that the samples remain representative. Import/export rules apply to this shipment.

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## Topic E.1: Use of Daily or Intermediate Wrap-ups

### Problem Statement

Can I have a daily or intermediate inspection wrap-up during a remote inspection?

### Recommendation

As in on-site inspections, the inspected site should ask the inspectors for daily (or frequent) wrap-ups during virtual inspections to clarify any outstanding issues and to discuss the following day's content. Doing so gives the inspected site time to prepare necessary documents and employees. Alternatively, inspectors could provide these wrap-ups at the beginning of the following day.

Some inspectors do not normally provide wrap-ups, but a regular wrap-up should always be requested since it will facilitate the ongoing inspection and help avoid potential misunderstandings regarding requested information. Inspectors also should promptly communicate any shift in focus to a new topic.

In desktop inspections, regular exchange of information on the status of the documentation review and potential concerns is equally important for the efficiency of the inspection process.

### Rationale

Risks related to miscommunication are higher during remote inspections than during on-site inspections due to the intrinsic limitations in communications. Regular wrap-ups held at a frequency congruent with the duration of the inspection can improve the efficiency and effectiveness of the overall process. These wrap-ups can allow the site to both identify and provide necessary clarifications and prepare documents and personnel for the next day's discussion. They also allow the inspector to ensure that the site clearly understands expectations and requests.

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# III. Remote Inspection Closing, Conclusion, and Response

## Topic A.1: Knowing an Inspection Has Ended

### Problem Statement

How do we know the inspection has ended?

### Recommendation

As mentioned in **Topic I.C.2 Planning the Agenda**, the inspector and inspected site should discuss the starting and ending dates for the inspection early in the process.

Remote inspections should end with a closing meeting via video communication platform, teleconference, or other communication method. If the inspectors do not clearly state that the inspection has ended, the inspected site should ask the inspectors for confirmation.

### Rationale

For remote inspections, which intrinsically consist of iterative requests for information between personnel in different locations and for which the duration might not have been communicated in advance, it may not be obvious that the inspection is complete. Therefore, it is best to explicitly ask the inspectors.

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## Topic A.2: Inspection Report and Findings and Response

### Problem Statement

Will an inspection report be issued for a remote inspection? How will the inspection findings be communicated? Is the response process for a remote inspection different from that of an on-site inspection?

### Recommendation

The inspector's task, regardless of the tools used, is to confirm compliance. Thus, the issuance of a report and communication of findings in a remote inspection should not differ from that of an on-site inspection. The common practice is for inspectors to finalize any observations identified during the remote inspection in a written inspection report, along with a request that the manufacturing site provide an answer and a plan for CAPAs, including a timeframe for response. This report is likely to describe, at a minimum:

- The documentation that was reviewed and by whom
- Whether physical aspects of the facility were assessed, and the methods used
- Inspection outcome
- Rationale for the decision

The inspection report also should indicate the processes that could not be assessed remotely and whether they will be assessed on-site to allow for completion of the inspection in a hybrid format.

The report also may explain the need for an on-site inspection. This may be a standard practice for some health authorities following remote inspections. For other health authorities, a follow-up on-site inspection may be necessary only when the remote assessment is inconclusive in evaluating the site's compliance status.

The process to respond to remote inspection findings should be the same as in on-site inspections unless the health authority communicates a different procedure. During the closing meeting or equivalent, the inspectors should communicate the path forward for the inspection response process.

### Rationale

While the remote inspection process is slightly different from an on-site inspection, the documentation and communication of the inspection outcome in most cases will remain the same, with the presentation of any inspectional findings, issuance of an inspection report, and expectations of responses as appropriate.

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## Topic A.3: Issuance of GMP/GDP Certificate

### Problem Statement

If the outcome of the remote inspection is positive, will a GMP/GDP certificate be issued?

### Recommendation

If the outcome of the remote inspection is positive, a GMP or GDP certificate should be issued as part of the normal activity of the health authority, as per the applicable local and regional laws.

The certificate might specify that virtual tools were used to conduct the inspection, but this distinction may not be necessary as GMP/GDP certificates issued after an on-site inspection often do not include discussion of all tools used (e.g., site tour, document review prior to arrival on site).

### Rationale

While the assessment process per se is slightly different in remote inspections than in on-site inspections, the decision to release a GMP/GDP certificate, and the validity of the certificate, is based on the evidence of compliance demonstrated during the inspection.

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## Topic B.1: Frequency of Reinspection

### Problem Statement

Is the frequency of reinspection impacted by the type of inspection (remote or on-site)?

### Recommendation

The frequency of reinspection depends on multiple parameters, including criticality of the product, complexity of the process, compliance history of the facility, regulatory resources, geographic location of the facility, and format of the inspection (e.g., on-site, desktop, or virtual). This determination will be guided by the health authority's risk-based approach. The next inspection following a remote inspection may occur at a reduced time interval due to the inherent limitations of a remote inspection as compared to on-site inspections.

### Rationale

Technology can help in performing a remote inspection that is adequate to allow the inspector to make an informed decision about the site's compliance status and/or compliance with the content in a regulatory submission. Even so, intrinsic limitations in a remote inspection may present challenges in accessing all the necessary information. Inspectorates are likely to take this into consideration when determining the frequency for the next inspection.

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# IV. References

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2. Pharmaceutical Inspection Convention/Cooperation Scheme (PIC/S). *Guidance: GMP Inspection Reliance*; PIC/S: Geneva, 2018.

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# V. Relevant Global Guidances

Australian Therapeutic Goods Administration, *TGA Expectations for Overseas Manufacturing Sites Hosting Remote Inspections during the COVID-19 Pandemic*. TGA: Woden ACT, Australia, 2020. https://www.tga.gov.au/tga-expectations-overseas-manufacturing-sites-hosting-remote-inspections-during-covid-19-pandemic.

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European Medicines Agency, *Guidance related to GMP/GDP and PMF Distant Assessments*. EMA: Brussels, 2020. https://www.ema.europa.eu/en/documents/scientific-guideline/guidance-related-gmp/gdp-pmf-distant-assessments_en.pdf.

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Vladislav Shestakov and Elizabeth Meyers, Regulator Develops Remote Inspection Process Due to Pandemic. *PDA Letter*, PDA, Inc.: Bethesda, Md., 2020. https://www.pda.org/pda-letter-portal/home/full-article/regulator-develops-remote-inspection-process-due-to-pandemic.

U.S. Food and Drug Administration, *Guidance for Industry: Remote Interactive Evaluations of Drug Manufacturing and Bioresearch Monitoring Facilities During the COVID-19 Public Health Emergency*. U.S. Department of Health and Human Services: Silver Spring, Md., 2021; 13 pp. https://www.fda.gov/media/147582/download.

U.S. Food and Drug Administration, Sec. 5.3.8.3.2, Electronic Databases and Queries. In *FDA Investigations Operations Manual*, U.S. Department of Health and Human Services: Silver Spring, Md., 2020; p 5-46. https://www.fda.gov/media/76769/download.

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# VII. Appendix 1: List of Quality Documents to be Reviewed During Inspection

The following list describes documents that may be requested by an inspector, although requests will vary depending on the scope of the inspection and the preferences of the inspectorate. This list is not exhaustive and is not intended to be a recommendation that all of the listed documents will need to be provided for any particular inspection.

**Table A1-1** Documents That May Be Requested by an Inspector

| No. | Requested Document |
|---|---|
| 1 | Site general presentation (including size, organization, and responsibilities) |
| 2 | Copy of site GMP certificate (issued by local authority) |
| 3 | Corrective action/preventive action for observations from the previous inspection |
| 4 | Description of the product(s) production flow – name, address, and manufacturing stages |
| 5 | Quality management review standard operating procedure (SOP) |
| 6 | Most recent Annual Product Review of the product(s) |
| 7 | Procedure for product release. |
| 8 | Procedure for change control management |
| 9 | Procedure for deviation management |
| 10 | Audit details (including audit plan and supplier audit plan) |
| 11 | Self-inspection procedure |
| 12 | Procedure for complaints management |
| 13 | Procedure for recall management |
| 14 | List of approved suppliers of materials for product(s) in scope of the inspection |
| 15 | Job descriptions for key personnel: Head of Manufacturing, Head of Quality Control |
| 16 | Detailed layout of manufacturing facility with indication of rooms grades, personnel & material flows, |
| 17 | Detailed layout of quality control laboratories |
| 18 | Copy of Master Batch Record for product(s) in scope of the inspection |
| 19 | In-process controls of manufacturing process |
| 20 | Report and protocol of manufacturing line qualification used for manufacturing of product(s) in scope of the inspection |
| 21 | Specification for finished drug product manufacturing product(s) in scope of the inspection |
| 22 | Specification for labeled drug product (before packaging) manufacturing product(s) in scope of the inspection |
| 23 | Stability study report for product(s) in scope of the inspection including data along all product shelf life. Procedure for products stability study. |
| 24 | Analytical methods validation (or methods transfer report) for product(s) in scope of the inspection (methods will be selected after specification receiving) |
| 25 | Procedure for out-of-specification management. |
| 26 | Reprocessed or reworked batches list |
| 27 | Environmental Monitoring procedure, locations, results, and trends (including monitoring of the utilities) |
| 28 | Site Master File |
| 29 | Aseptic Process Simulations |
| 30 | Cleaning & cleaning validation |

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| | |
|---|---|
| 31 | Equipment validation/qualification, calibration, and maintenance |
| 32 | Process validation |
| 33 | Incoming material and supplier management |
| 34 | Risk management |
| 35 | Shipping details and SOP |
| 36 | Design history files or drug master file |
| 37 | Contamination control strategy |


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# VIII. Appendix 2: *2021 PDA Research: Global Industry Responses to Remote Regulatory Inspections*

In 2020, global health authorities increased their use of remote inspections of pharmaceutical and biopharmaceutical manufacturers due to COVID-19 precautions. Manufacturers and health authorities alike worked rapidly to adapt on-site inspection practices to the virtual environment and to overcome related challenges.

PDA conducted this survey to better understand industry's response to these changes, as well as potential future needs. To gain these insights, PDA sought the input of global pharmaceutical and biopharmaceutical employees who help develop company policies and procedures relating to regulatory inspections, interact with regulatory inspectors, and/or provide support during regulatory inspections. The survey was targeted at individuals who work for a company that manufactures finished products for human use, active pharmaceutical ingredients (APIs), or excipients.

## Survey Methodology

The survey on *Global Industry Approaches to Remote Regulatory Inspections* was open for participation from November 9 through November 24, 2020.

A total of 59 respondents completed the survey. Survey respondents are experts involved in regulatory inspections within their current companies, whether from the regulatory affairs, manufacturing, quality, compliance, or other functional perspective. The survey results include a good balance between drug product manufacturers and API or excipient manufacturers, as well as responses from sites of varying sizes. Respondents included industry professionals located in North America, Europe, Asia/Pacific, Africa, and the Middle East. Latin America was not represented among respondents.

The survey used skip logic to ask follow-up questions to selected subsets of respondents. For instance, Questions 8 through 13 were asked only of those who said in response to Question 7 that they had participated in or supported a remote regulatory inspection or assessment in 2020. As a result of this skip logic, the number of respondents varies depending on the question asked.

For numerous questions, respondents could select as many options as were applicable. These questions are denoted by the instruction to "Select all that apply." The percentages shown in the responses to these questions represent the percent of *respondents* rather than percent of total responses. The percentages for these questions will not total 100.

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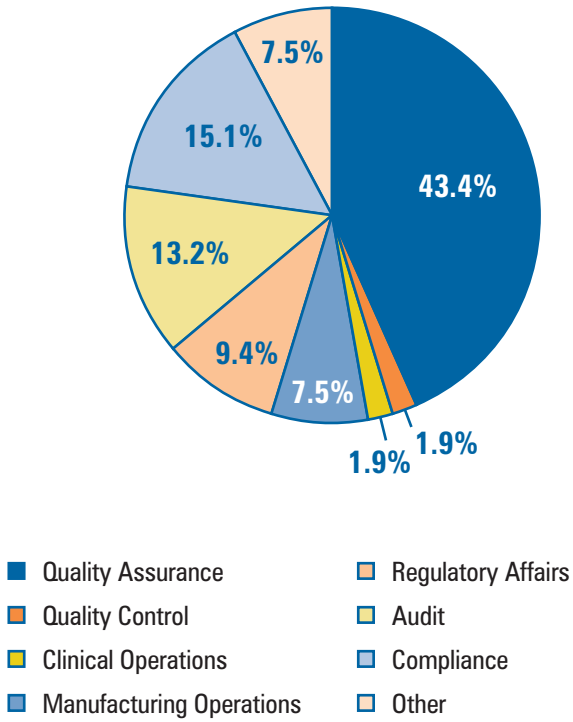
## Survey Results

1. *Which of these best describes your primary role?*

| Answer | Count | % |
|---|---|---|
| Quality Assurance | 23 | 43.4% |
| Quality Control | 1 | 1.9% |
| Clinical Operations | 1 | 1.9% |
| Manufacturing Operations | 4 | 7.5% |
| Regulatory Affairs | 5 | 9.4% |
| Audit | 7 | 13.2% |
| Compliance | 8 | 15.1% |
| Other | 4 | 7.5% |
| ***Answered*** | ***53*** | |

Other responses:
- Medical Devices
- I am consultant and work with all of the above
- Vaccines
- Medical Devices (×2)
- Cell Therapy

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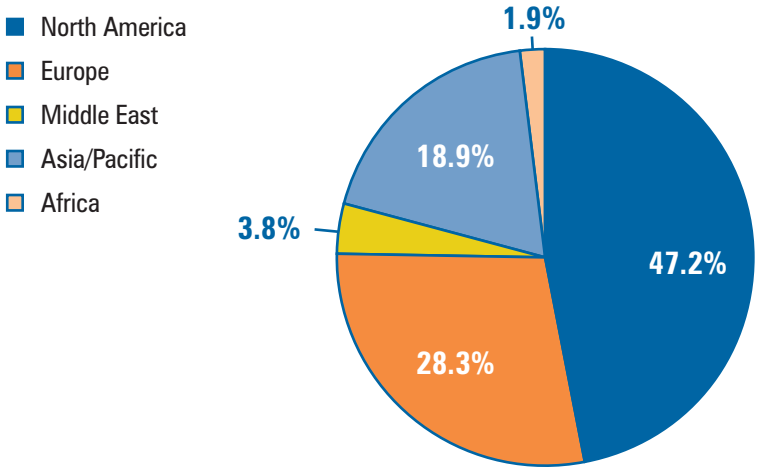
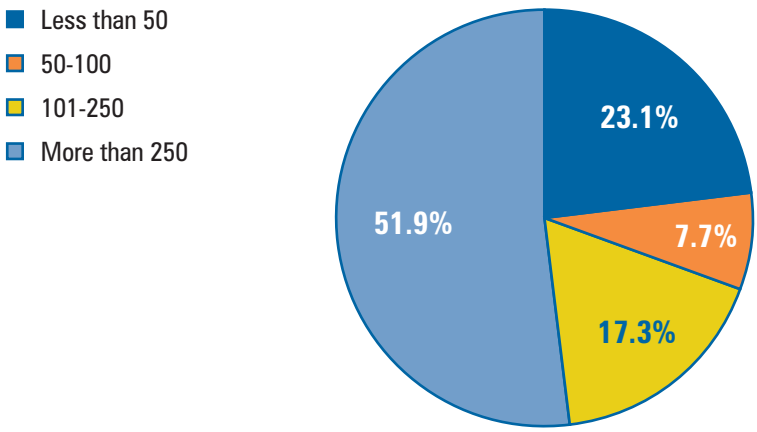
2. *In what region are you located?*

| Answer | Count | % |
|---|---|---|
| North America | 25 | 47.2% |
| Europe | 15 | 28.3% |
| Middle East | 2 | 3.8% |
| Asia/Pacific | 10 | 18.9% |
| Africa | 1 | 1.9% |
| ***Answered*** | ***53*** | |




3. *How many employees are at your location?*

| Answer | Count | % |
|---|---|---|
| Less than 50 | 12 | 23.1% |
| 50-100 | 4 | 7.7% |
| 101-250 | 9 | 17.3% |
| More than 250 | 27 | 51.9% |
| ***Answered*** | ***52*** | |





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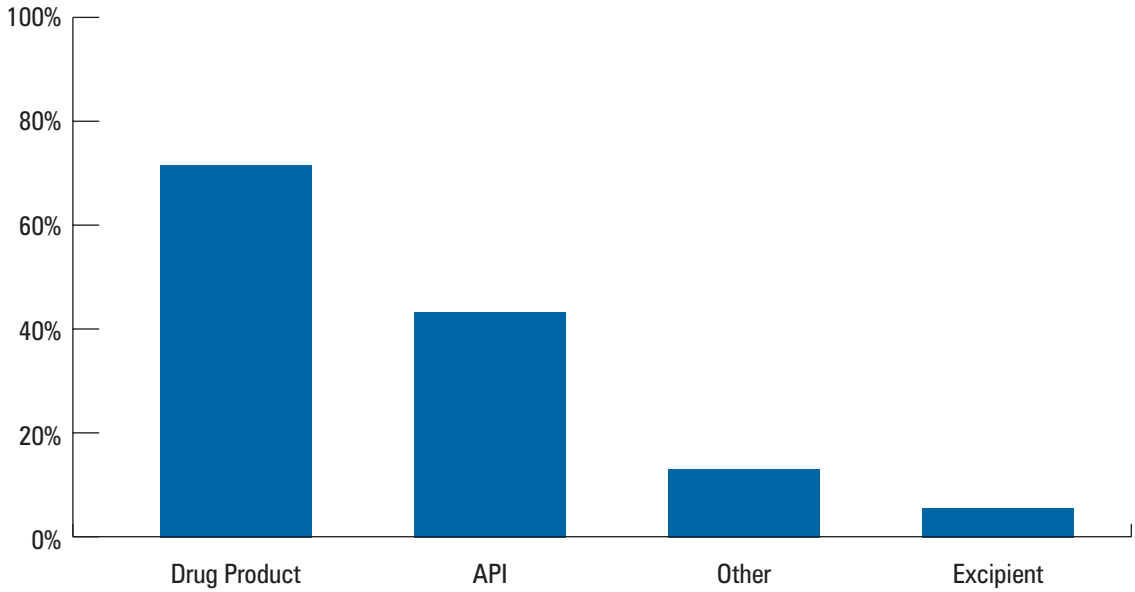
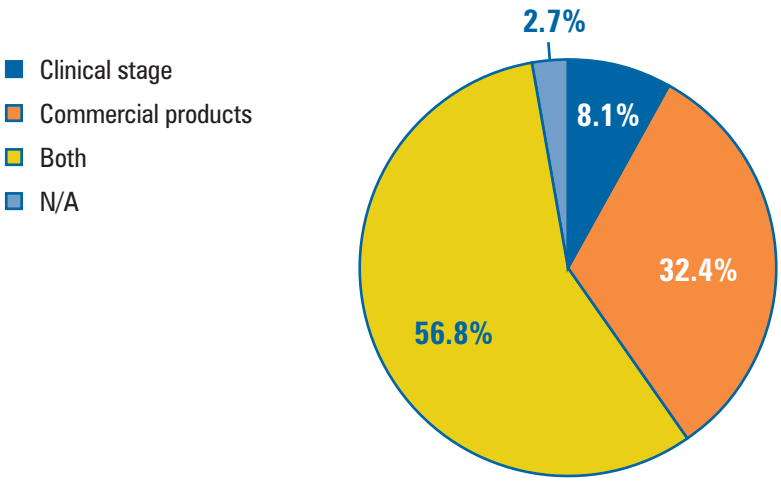
4. *What types of products do you manufacture/develop at your site? Select all that apply.*

| Answer | Count | % |
|---|---|---|
| Drug product | 38 | 71.7% |
| Active pharmaceutical ingredient (API) or formulated bulk | 23 | 43.4% |
| Other | 7 | 13.2% |
| Excipient | 3 | 5.7% |
| ***Answered*** | ***53*** | |




5. *If you manufacture/develop drug products, do you manufacture/develop for:*

| Answer | Count | % |
|---|---|---|
| Clinical stage | 3 | 8.1% |
| Commercial products | 12 | 32.4% |
| Both | 21 | 56.8% |
| N/A | 1 | 2.7% |
| ***Answered*** | ***37*** | |





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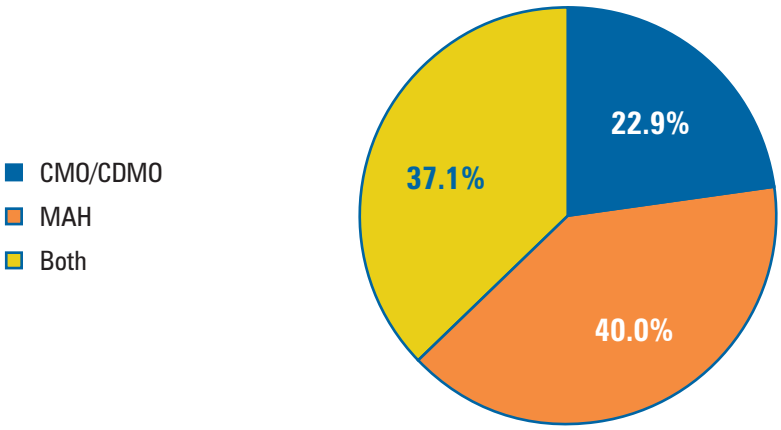
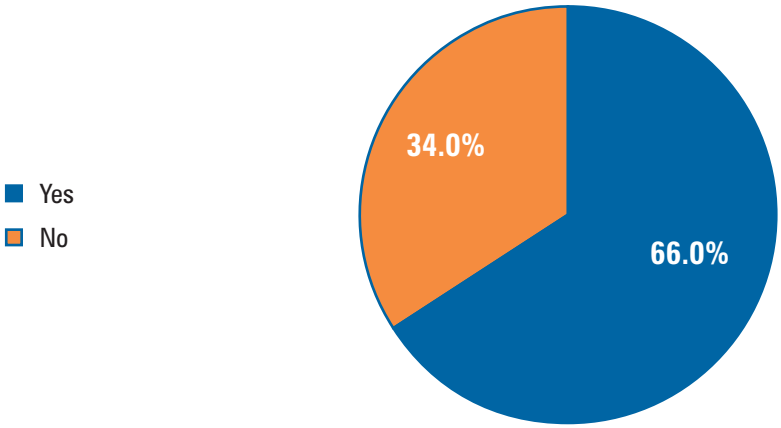
6. *If you manufacture/develop drug products, do you manufacture/develop:*

| Answer | Count | % |
|---|---|---|
| As a contract manufacturing organization (CMO) / contract development and manufacturing organization (CDMO) | 8 | 22.9% |
| As a marketing authorization holder (MAH) | 14 | 40.0% |
| Both | 13 | 37.1% |
| ***Answered*** | ***38*** | |




7. *In 2020, have you participated in or supported a regulatory inspection or assessment conducted remotely?*

| Answer | Count | % |
|---|---|---|
| Yes | 35 | 66.0% |
| No | 18 | 34.0% |
| ***Answered*** | ***53*** | |





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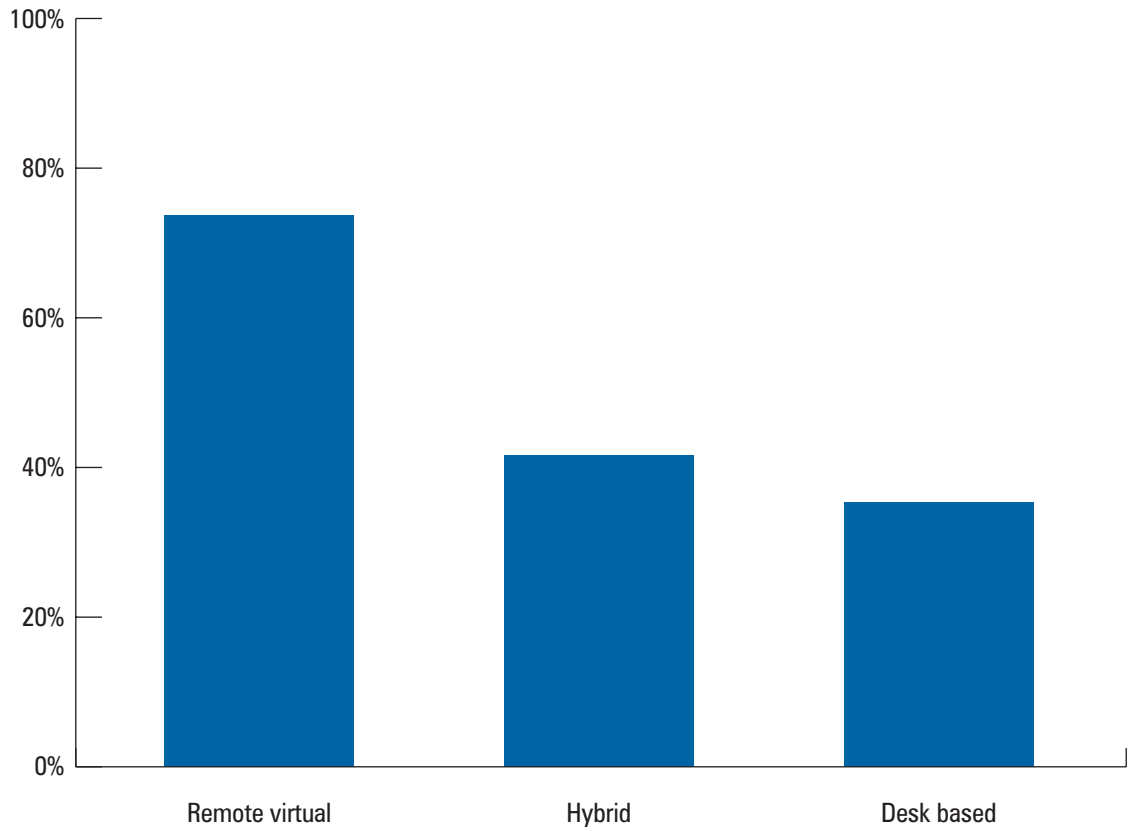
8. *If you answered Yes to Question 7,* what *was the format of the remote regulatory inspection(s)/ assessment(s) that you participated in/supported? Select all that apply.*

| Answer | Count | % |
|---|---|---|
| Remote virtual, with discussions/interviews occurring via teleconference or videoconference | 23 | 74.2% |
| Hybrid, with an inspector spending some time in person at your facility, and some of the inspection/assessment conducted remotely or virtually | 13 | 41.9% |
| Desk-based: records review only, with no live discussions or interviews | 11 | 35.5% |
| ***Answered*** | ***31*** | |





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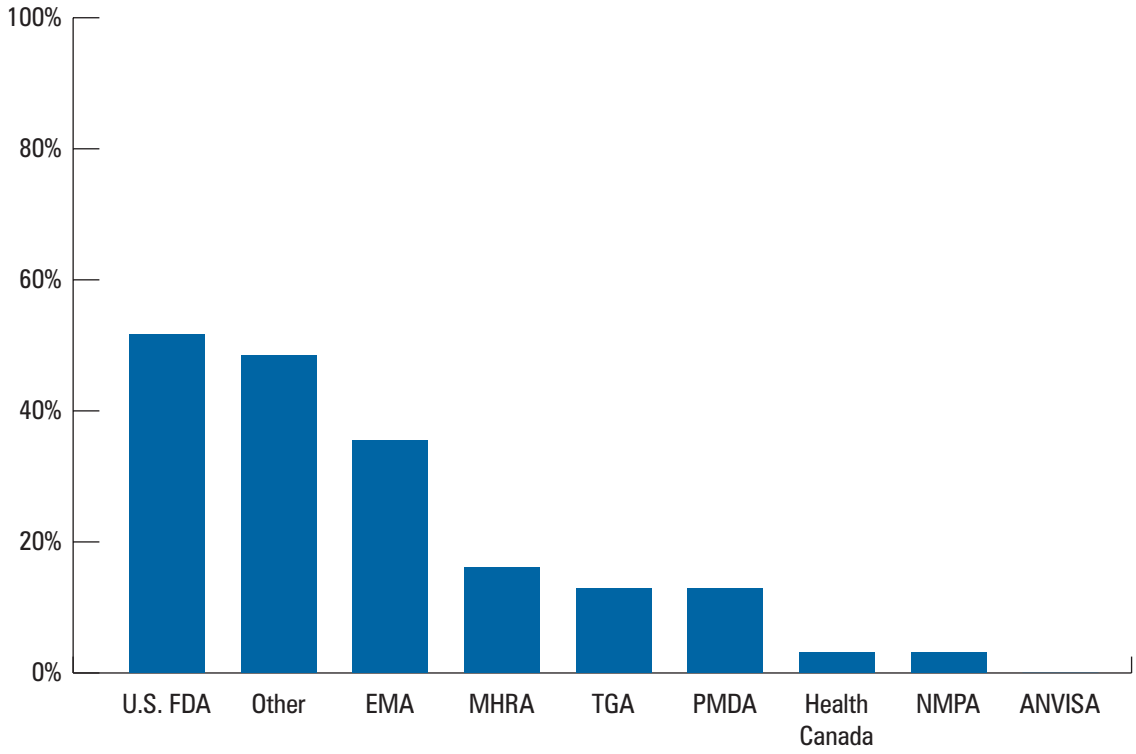
9. *Which regulatory authority/authorities conducted the remote inspection(s)/assessment(s)? Select all that apply.*

| Answer | Count | % |
|---|---|---|
| U.S. FDA | 16 | 51.6% |
| Other | 15 | 48.4% |
| EMA (EU) | 11 | 35.5% |
| MHRA (UK) | 5 | 16.1% |
| TGA (Australia) | 4 | 12.9% |
| PMDA (Japan) | 4 | 12.9% |
| Health Canada | 1 | 3.2% |
| NMPA (China) | 1 | 3.2% |
| Anvisa (Brazil) | 0 | 0.0% |
| ***Answered*** | ***31*** | |

Other responses:
- ANSM France
- notified body
- Russia (×4)
- Russia (MoIT) (×2)
- Swiss Medic
- Turkey; Russia (most)

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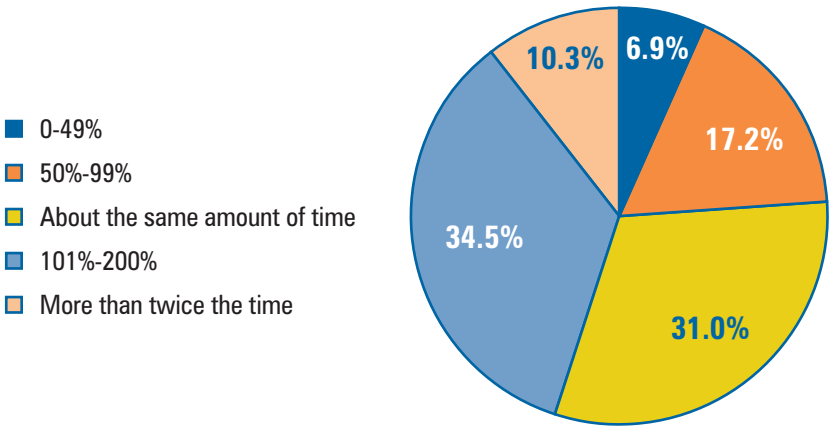
10. *Did the remote inspection(s)/assessment(s) take:*

| Answer | Count | % |
|---|---|---|
| Less than half (0 – 49%) of the time that an in-person inspection would have taken. (It was much shorter than an in-person inspection.) | 2 | 6.9% |
| 50% – 99% of the time that an in-person inspection would have taken. (It was somewhat shorter than an in-person inspection.) | 5 | 17.2% |
| About the same amount of time as an in-person inspection | 9 | 31.0% |
| 101% – 200% of the time than an inspection would have taken. (It was somewhat longer than an in-person inspection.) | 10 | 34.5% |
| More than twice the time that an in-person inspection would have taken. (It was significantly longer than an in-person inspection.) | 3 | 10.3% |
| ***Answered*** | ***29*** | |

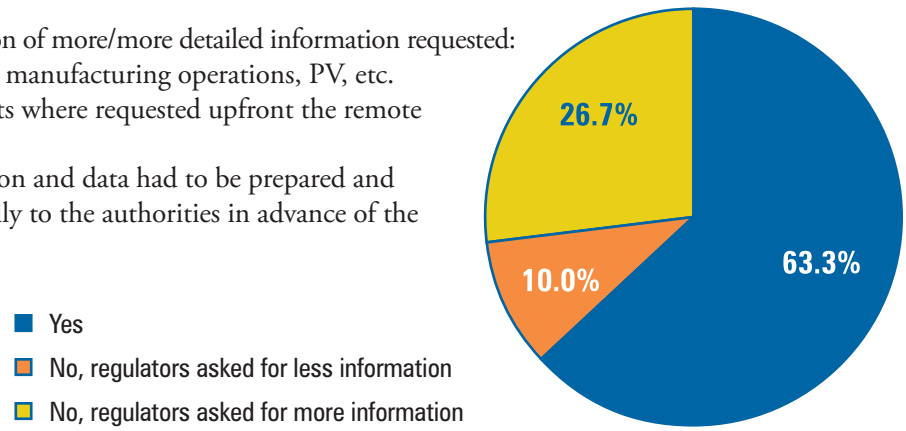



11. *In the remote inspections/assessments, were the requests/demands made by the regulators equivalent to those that would be made in an in-person inspection??*

| Answer | Count | % |
|---|---|---|
| Yes, the regulators made basically the same requests/demands that we would expect in an in-person inspection | 19 | 63.3% |
| No, the regulators asked for less information than they would in an in-person inspection | 3 | 10.0% |
| No, the regulators asked for more information or more detailed information than in an in-person inspection. (Please describe briefly.) | 8 | 26.7% |
| ***Answered*** | ***30*** | |

Additional description of more/more detailed information requested:

- More details on manufacturing operations, PV, etc.
- More documents where requested upfront the remote inspection
- More information and data had to be prepared and sent electronically to the authorities in advance of the inspection.

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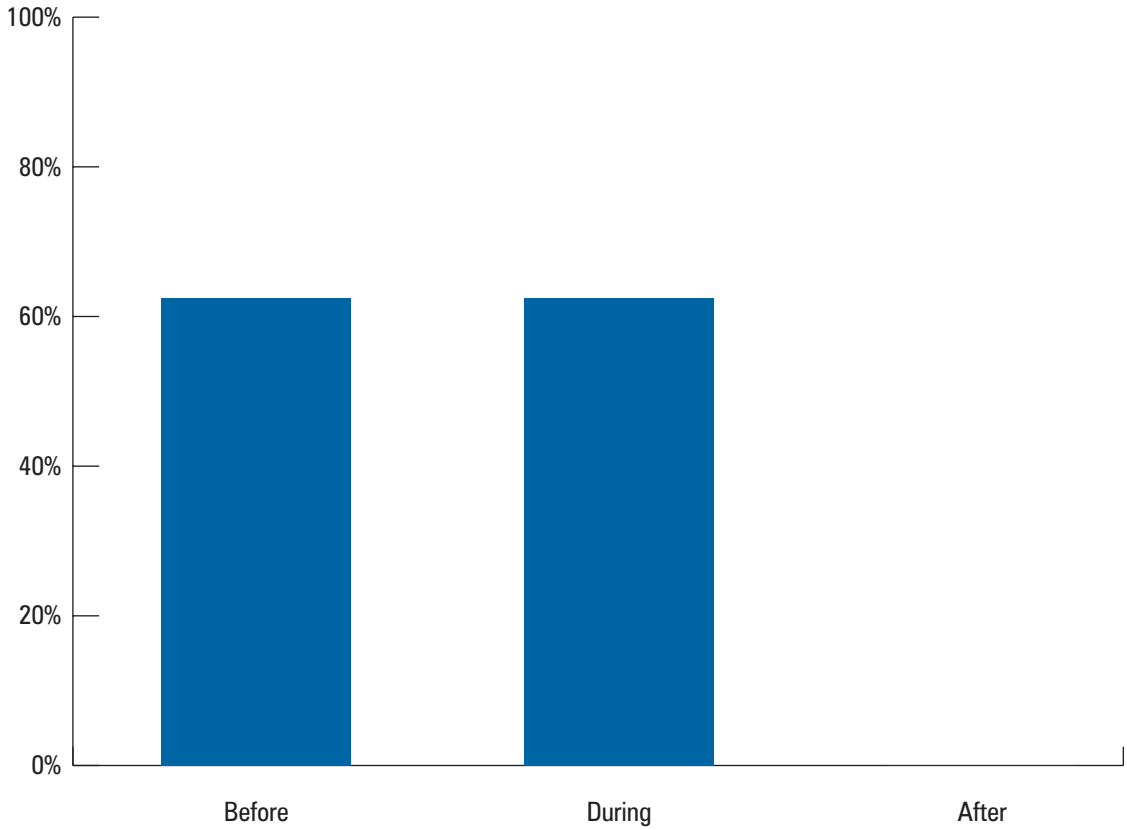
12. *If you answered "No, the regulators asked for more information or more detailed information" in Question 11, at what stage of the remote inspection did the regulatory authority request/demand this additional or more detailed information? Select all that apply.*

| Answer | Count | % |
|---|---|---|
| Before the inspection began | 5 | 62.5% |
| During the inspection | 5 | 62.5% |
| After the inspection concluded | 0 | 0.0% |
| ***Answered*** | ***8*** | |

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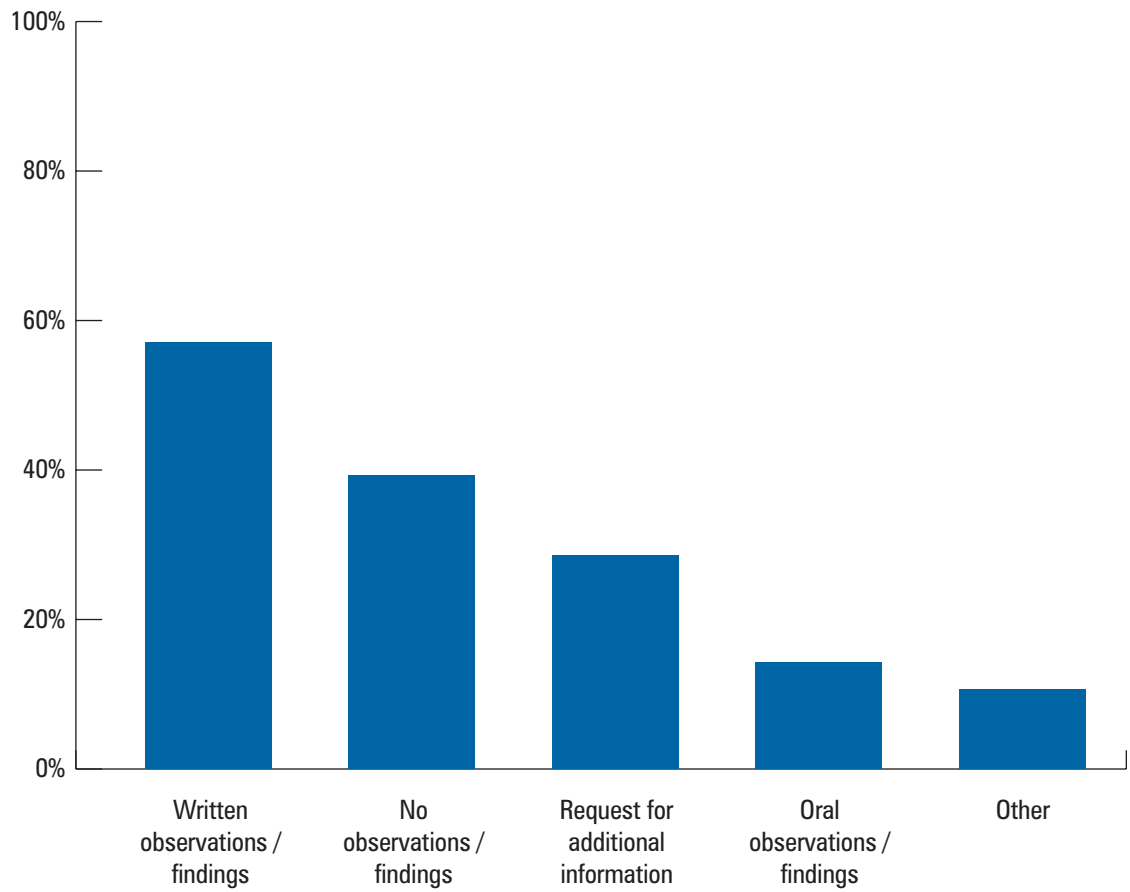
13. *What was the outcome of the remote inspection? Select all that apply.*

| Answer | Count | % |
|---|---|---|
| Written observations / findings | 16 | 57.1% |
| No observations / findings | 11 | 39.3% |
| Request for additional information | 8 | 28.6% |
| Oral observations / findings | 4 | 14.3% |
| Other | 3 | 10.7% |
| ***Answered*** | ***28*** | |

Other responses:
- Approval
- One day on site inspection

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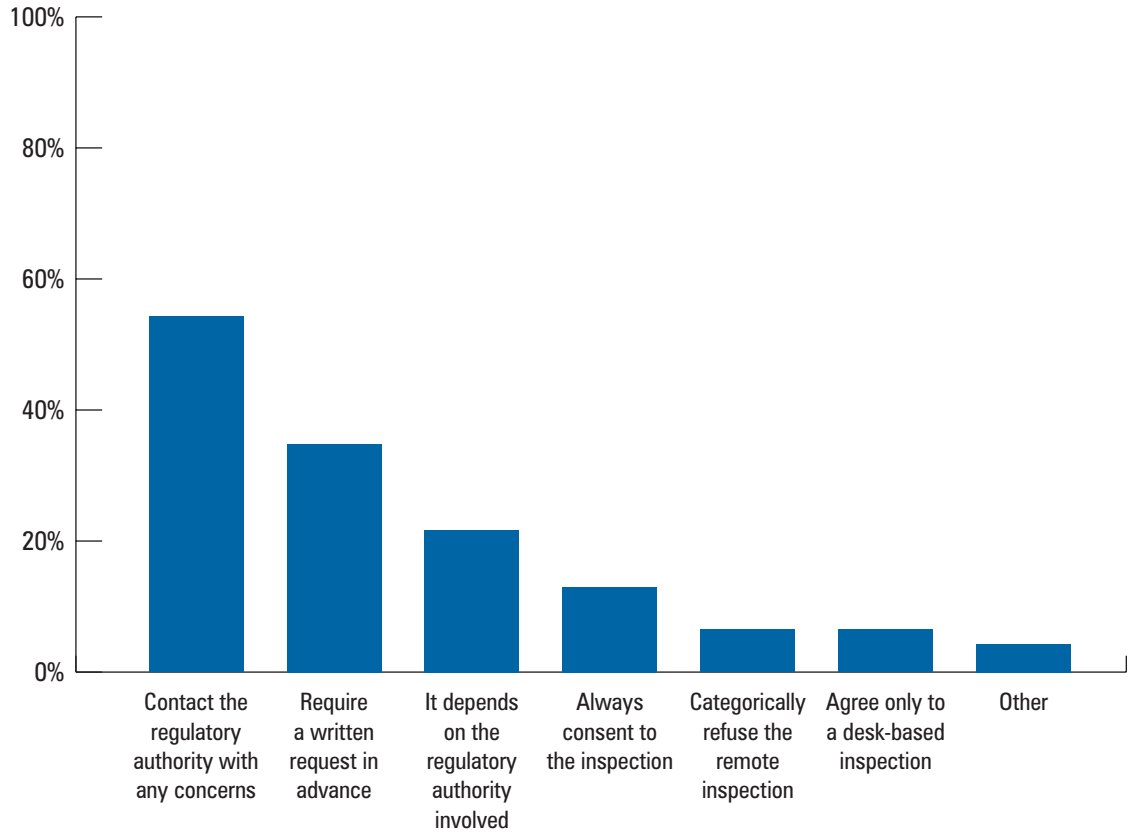
14. *If your company had concerns about the legitimacy of an inspector that asks to conduct a remote inspection of your site, which of these answers best describes how you would respond? Select all that apply.*

| Answer | Count | % |
|---|---|---|
| We would contact the regulatory authority with any concerns about the inspector's legitimacy | 25 | 54.3% |
| We would require a written request in advance and would authenticate the source of the written request | 16 | 34.8% |
| It depends on the regulatory authority involved | 10 | 21.7% |
| We would always consent to the inspection, even if we had concerns about the legitimacy of the inspector | 6 | 13.0% |
| We would categorically refuse the remote inspection if we were concerned about the inspector's legitimacy | 3 | 6.5% |
| We would agree only to a desk-based inspection, limited to records | 3 | 6.5% |
| Other (please specify) | 2 | 4.3% |
| ***Answered*** | ***46*** | |

Other responses:

- Never had this concern. Always received a copy of the formal inspectorate letter notification
- The purpose of an inspection is to check compliance. No law requests how to do that. The inspector must be confident of compliance.

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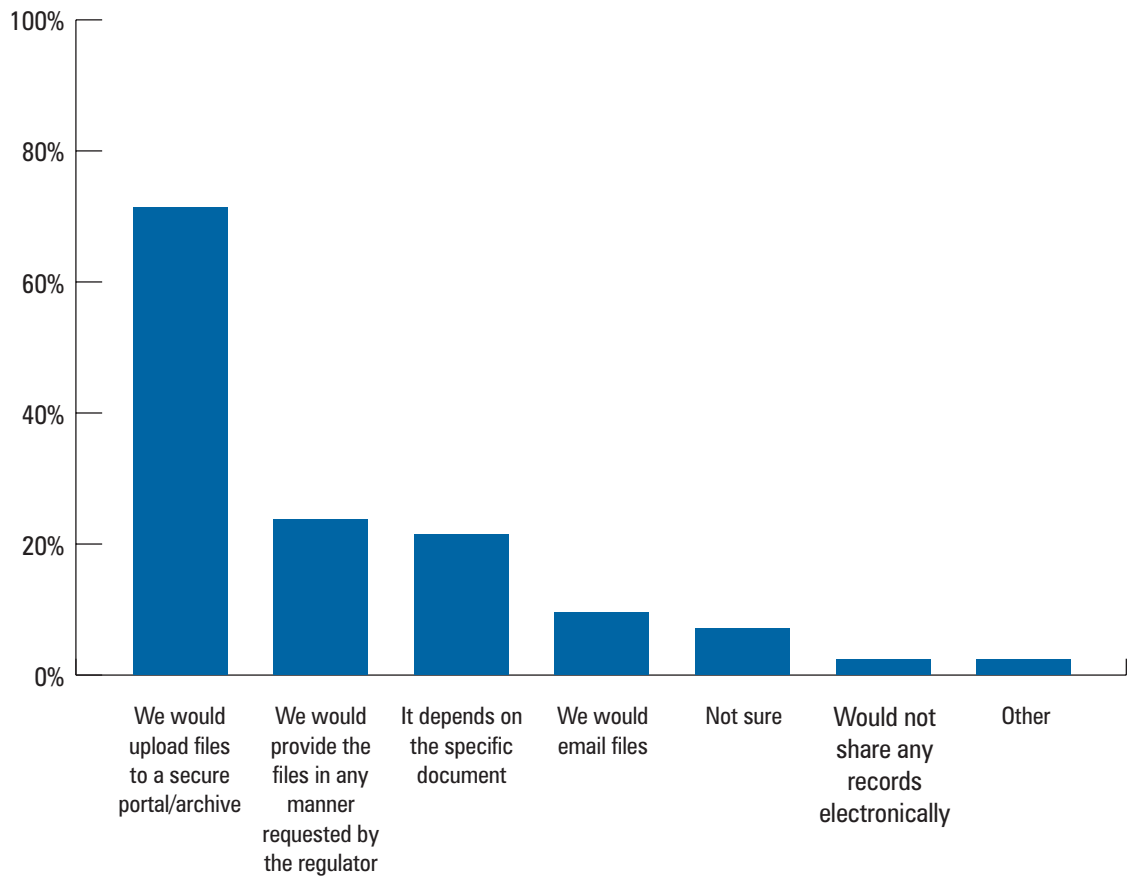
15. *During an inspection conducted remotely, how would you provide records/documents to the regulatory authority? Select all that apply.*

| Answer | Count | % |
|---|---|---|
| We would upload files to a secure portal/archive | 30 | 71.4% |
| We would provide the files in any manner requested by the regulator | 10 | 23.8% |
| It depends on the specific document. We would apply greater protections to some documents than to others | 9 | 21.4% |
| We would email files | 4 | 9.5% |
| Not sure | 3 | 7.1% |
| We would not share any records electronically. We would send any necessary records by courier or mail (e.g., in hard copy or on USB drive) | 1 | 2.4% |
| Other (please specify) | 1 | 2.4% |
| ***Answered*** | ***42*** | |





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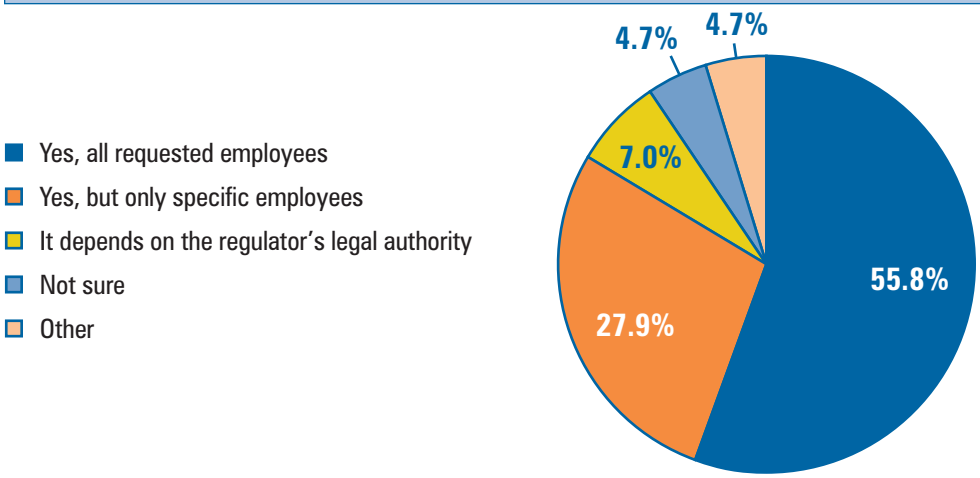
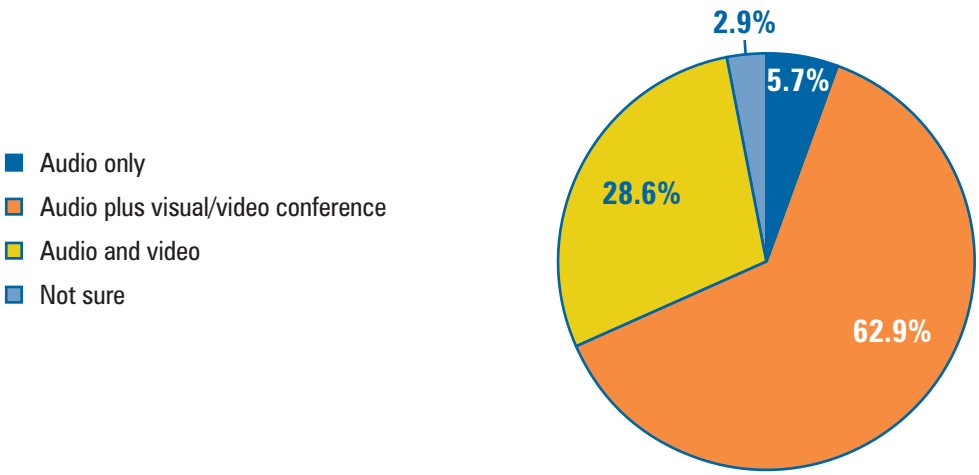
16. *Would you agree to make employees available for remote (audio and/or video) interviews by the investigator?*

| Answer | Count | % |
|---|---|---|
| Yes, all requested employees | 24 | 55.8% |
| Yes, but only specific employees | 12 | 27.9% |
| It depends on the regulator's legal authority | 3 | 7.0% |
| Not sure | 2 | 4.7% |
| Other (please specify) | 2 | 4.7% |
| ***Answered*** | ***43*** | |




17. *If you would allow interviews, in what formats would you allow interviews?*

| Answer | Count | % |
|---|---|---|
| Audio only, no visual/video of the person being interviewed | 2 | 5.7% |
| Audio plus visual/video conference, provided that the agency officials are also on video | 22 | 62.9% |
| Audio and video, without limitation | 10 | 28.6% |
| Not sure | 1 | 2.9% |
| ***Answered*** | ***35*** | |





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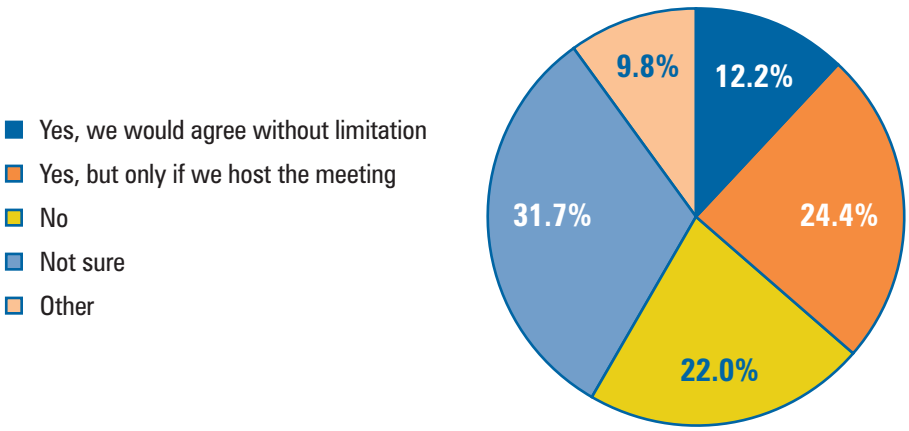
18. *During a remote inspection conducted by videoconference (e.g. Zoom, Microsoft Teams, GoToMeeting), would you allow the regulatory authority to record the proceedings?*

| Answer | Count | % |
|---|---|---|
| Yes, we would agree without limitation | 5 | 12.2% |
| Yes, but only if we (the inspected site) host the meeting on a platform we control that has been approved by our IT Security Team | 10 | 24.4% |
| No | 9 | 22.0% |
| Not sure | 13 | 31.7% |
| Other (please specify) | 4 | 9.8% |
| ***Answered*** | ***41*** | |

Other responses:

- Audio only by agreement
- Never been requested.
- Similar to what counts of on-site inspections
- The agreement to record all or part of the inspection has to be taken in advance of the virtual inspection

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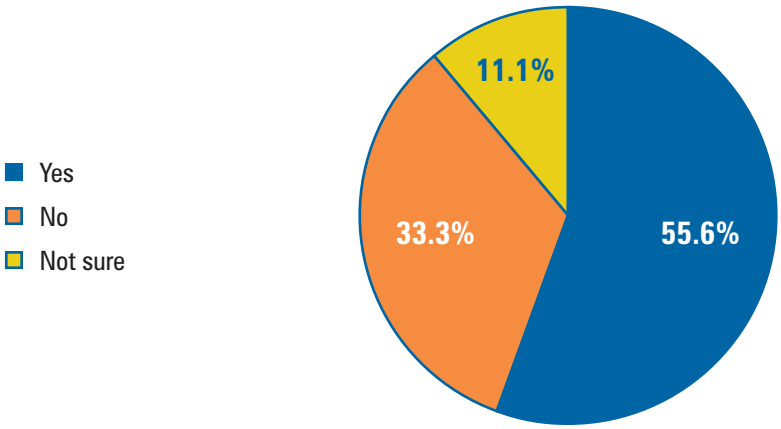
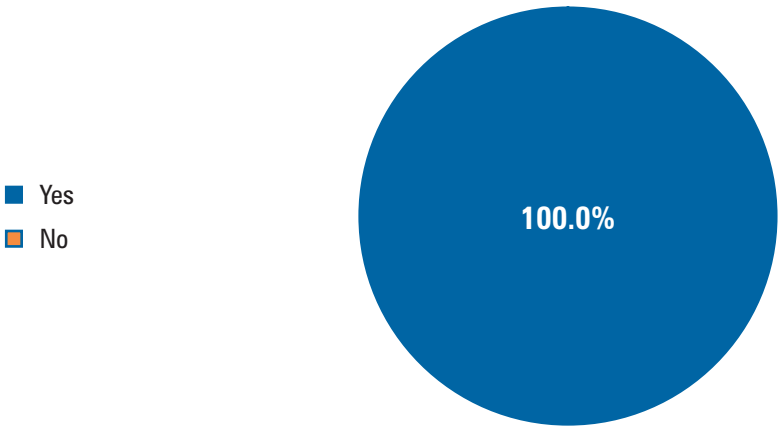
19. *If you answered No to Question 18, would you require written assurance that the regulator is not recording the sessions?*

| Answer | Count | % |
|---|---|---|
| Yes | 5 | 55.6% |
| No | 3 | 33.3% |
| Not sure | 1 | 11.1% |
| ***Answered*** | ***9*** | |




20. *If you answered Yes to Question 19, would you be willing to provide reciprocal assurance (that you are not recording the sessions) to the regulator?*

| Answer | Count | % |
|---|---|---|
| Yes | 5 | 100.0% |
| No | 0 | 0.0% |
| ***Answered*** | ***5*** | |

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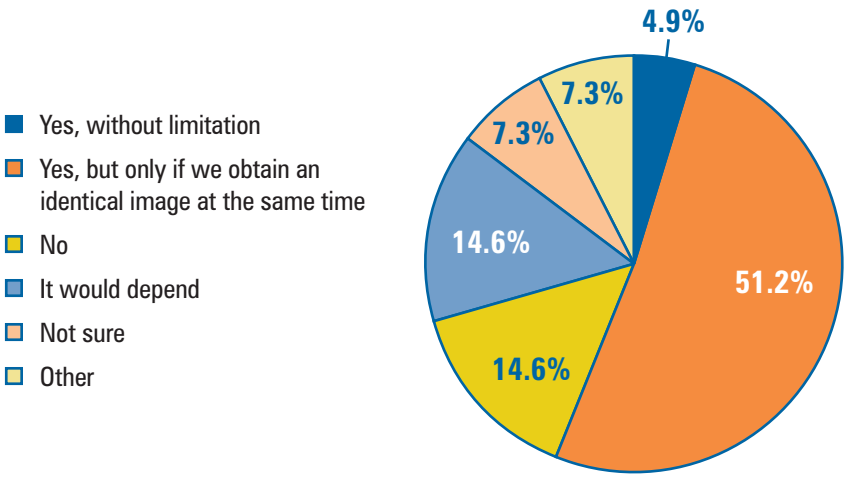
21. *If asked during a remote inspection, would you allow a regulator to capture a still image (photo, screenshot, or other) of a condition or practice within your facility?*

| Answer | Count | % |
|---|---|---|
| Yes, without limitation | 2 | 4.9% |
| Yes, but only if we obtain an identical image at the same time | 21 | 51.2% |
| No | 6 | 14.6% |
| It would depend on what the desired image depicted | 6 | 14.6% |
| Not sure | 3 | 7.3% |
| Other (please specify) | 3 | 7.3% |
| ***Answered*** | ***41*** | |

Other responses:

- GVP inspections we would agree to provide the required image
- Same conditions apply as for on-site inspections
- Would depend on the desired image. If allowed, then image would be made by company and shared with inspector so both parties have identical image

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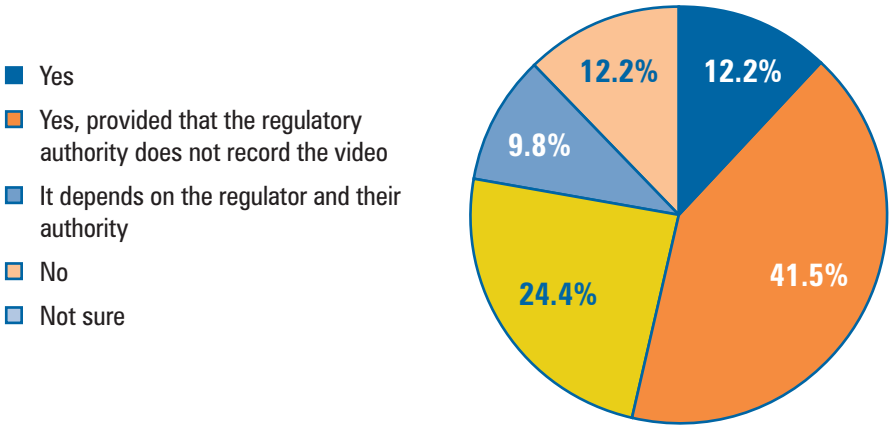
22. *If asked during a remote inspection, would you allow an employee to transmit live video from your manufacturing area, via the camera on a laptop, tablet, smartphone, or advanced technology such as smart glasses, to a regulator/inspector so that the regulator can observe conditions or practices within your facility?*

| Answer | Count | % |
|---|---|---|
| Yes | 5 | 12.2% |
| Yes, provided that the regulatory authority does not record the video | 17 | 41.5% |
| It depends on the regulator and their authority. We might allow it for some regulators and not others | 10 | 24.4% |
| No | 4 | 9.8% |
| Not sure | 5 | 12.2% |
| ***Answered*** | ***41*** | |

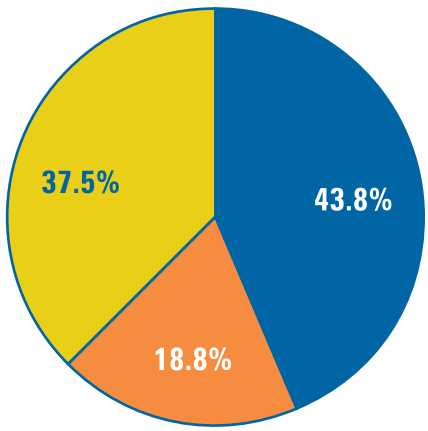



23. *If you answered "Yes, provided that the regulatory authority does not record the video" to Question 22, would you attempt to verify that a recording was not made?*

| Answer | Count | % |
|---|---|---|
| Yes. Please describe how you might verify: | 7 | 43.8% |
| No | 3 | 18.8% |
| Not sure | 6 | 37.5% |
| ***Answered*** | ***16*** | |

Descriptions of how respondents might verify:

- I would simply ask
- Through video technology indicators on recording or no recording
- Agreement
- We would agree to rules of engagement up front, but make no other attempt to verify

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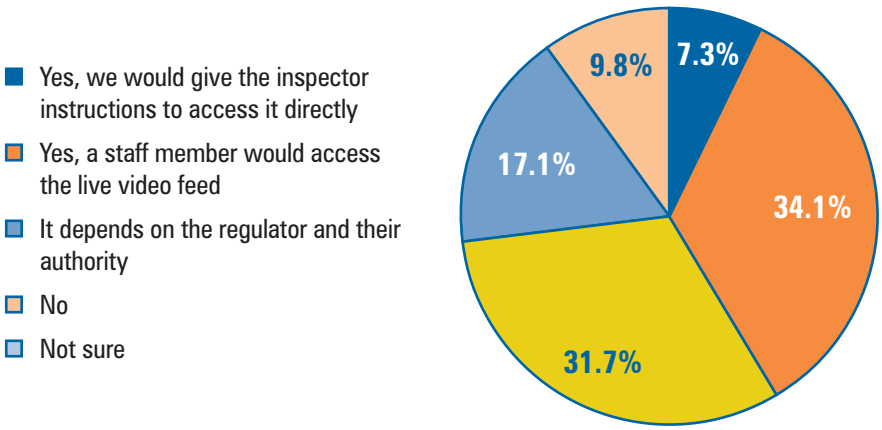
24. *If asked during a remote inspection, would you allow an inspector to view your own live video feed of your facility, for instance by viewing your CCTV footage or other internal camera feed?*

| Answer | Count | % |
|---|---|---|
| Yes, we would give the inspector instructions to access it directly | 3 | 7.3% |
| Yes, a staff member would access the live video feed and would share the images/video with the inspector (for instance through screen-sharing in a videoconference) | 14 | 34.1% |
| It depends on the regulator and their authority. We might allow it for some regulators and not others | 13 | 31.7% |
| No | 7 | 17.1% |
| Not sure | 4 | 9.8% |
| ***Answered*** | ***41*** | |

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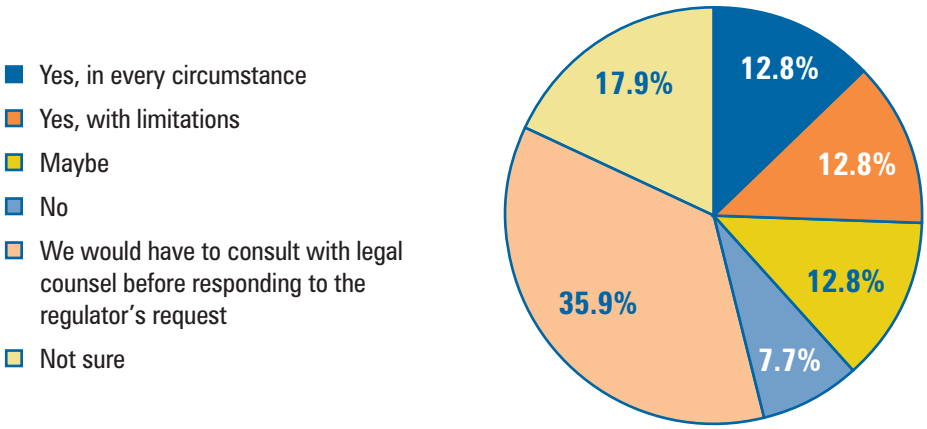
25. *If asked, would you provide a regulator with a physical sample of material (for example, foreign matter in process materials or finished product) as part of the remote inspection?*

| Answer | Count | % |
|---|---|---|
| Yes, in every circumstance | 5 | 12.8% |
| Yes, with limitations | 5 | 12.8% |
| Maybe. It depends on the regulatory authority. We might provide a sample to some regulatory authorities and not to others | 5 | 12.8% |
| No | 3 | 7.7% |
| We would have to consult with legal counsel before responding to the regulator's request | 14 | 35.9% |
| Not sure | 7 | 17.9% |
| ***Answered*** | ***39*** | |

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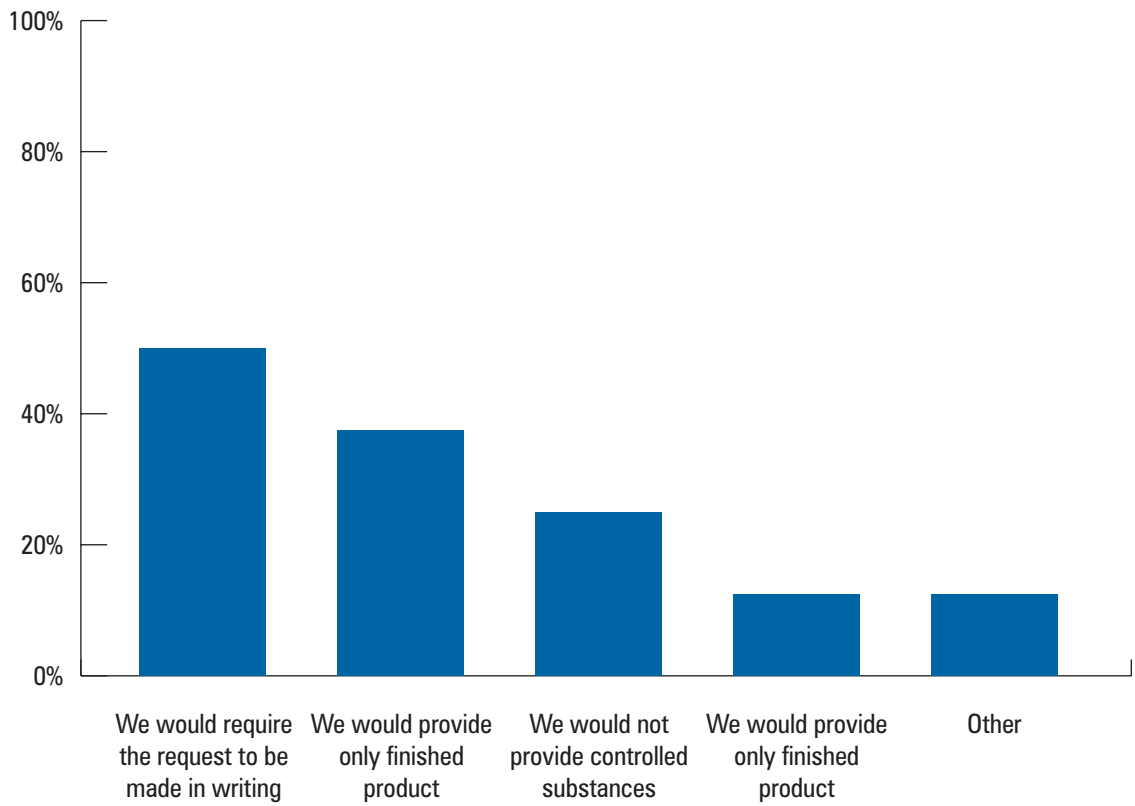
26. *If you answered "Yes, with limitations" to Question 25, what limitations would you impose? Select all that apply.*

| Answer | Count | % |
|---|---|---|
| We would require the request to be made in writing | 4 | 50.0% |
| We would provide only finished product that had been released for shipment per internal QA procedure | 3 | 37.5% |
| We would not provide controlled substances | 2 | 25.0% |
| We would provide only finished product | 1 | 12.5% |
| Other (please specify) | 1 | 12.5% |
| ***Answered*** | ***5*** | |

Other responses:

- According to the requirements applying in the law in the country the inspector comes from

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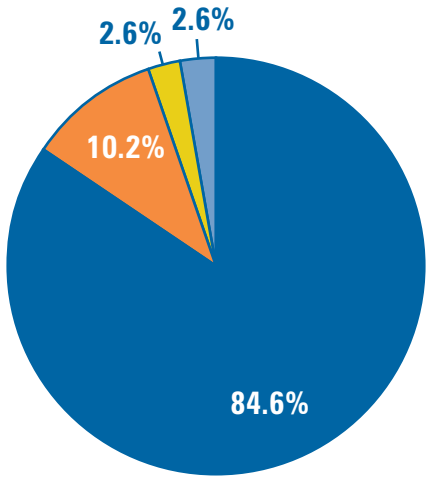
27. *At the conclusion of a remote inspection:*

| Answer | Count | % |
|---|---|---|
| We would participate in a remote exit conference without limitation | 33 | 84.6% |
| We would require that an exit conference and presentation of findings be conducted in person | 4 | 10.2% |
| We would only reply in writing to any written list of inspection observations | 1 | 2.6% |
| It depends on the regulatory authority involved | 1 | 2.6% |
| ***Answered*** | ***39*** | |

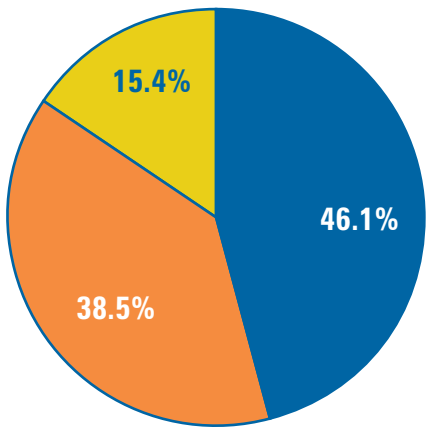



- We would participate in a remote exit conference without limitation
- We would require that an exit conference and presentation of findings be conducted in person
- We would only reply in writing to any written list of inspection observations
- It depends on the regulatory authority involved

28. *Have you begun changing SOPs and/or policies to proactively address remote regulatory inspections?*

| Answer | Count | % |
|---|---|---|
| Yes | 18 | 46.1% |
| No | 15 | 38.5% |
| We don't expect to change our SOPs | 6 | 15.4% |
| ***Answered*** | ***39*** | |

- Yes
- No
- We don't expect to change our SOPs





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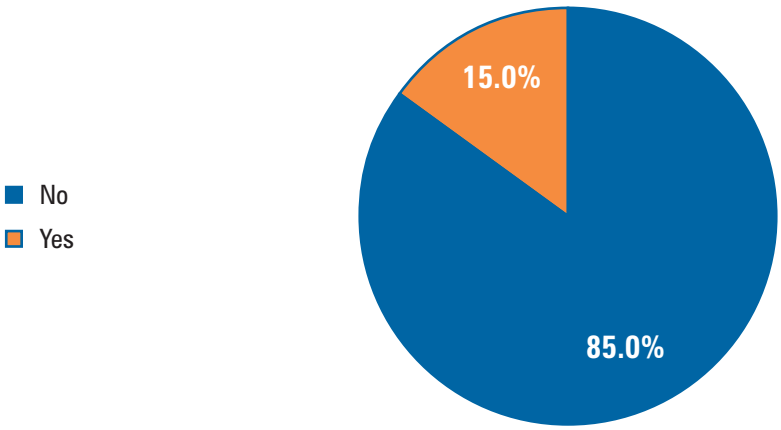
29. *Has your company identified questions or topics that are specific to remote inspections/assessments that that should be raised to legal counsel (in-house or outside counsel) before employees respond?*

| Answer | Count | % |
|---|---|---|
| No, the questions and topics are the same as in an in-person inspection | 34 | 85.0% |
| Yes | 6 | 15.0% |
| ***Answered*** | ***40*** | |




30. *If other aspects of remote regulatory inspections cause you concern, please describe them.*

Responses:

- 1) There should be a differentiation between initial inspection and re-inspection and inspector needs to be aware of it. 2) Inspectors might review other info about the site prior to the inspection and the site should be aware of it.
- Additional document/information requests during virtual inspection follow up can extend over weeks; there should be a time limitation by when requests are accepted
- Be clear on the terminology. It seems 'remote inspections' refers to 'paper based + 'virtual inspections'
- Confidentiality
- Increased frequency of inspections due to no travel time lost and flexibility of remote inspecting
- It has put strain on our system, and we see lack of motivation in employees to sit in front of the system for long durations
- Miscommunication or misinterpretation with limited information must be avoided or clarity must be established before conclusion
- Possible request to use teleconferencing software (e.g. Zoom, GoToMeeting) that are not among those that are company-approved (Microsoft Teams, WebEx)
- Presentation of trade secret documents
- Remote inspections take more time to cover the same amount of ground as an in-person inspection, and require a well-prepared backroom to ensure that requests are met quickly and accurately. Greater potential for misunderstandings.
- Technical limitations could provide a distorted "picture"/view on a number of topics throughout a remote inspection
- Technology limitations – they can be falsified easily, which breeds mistrust.

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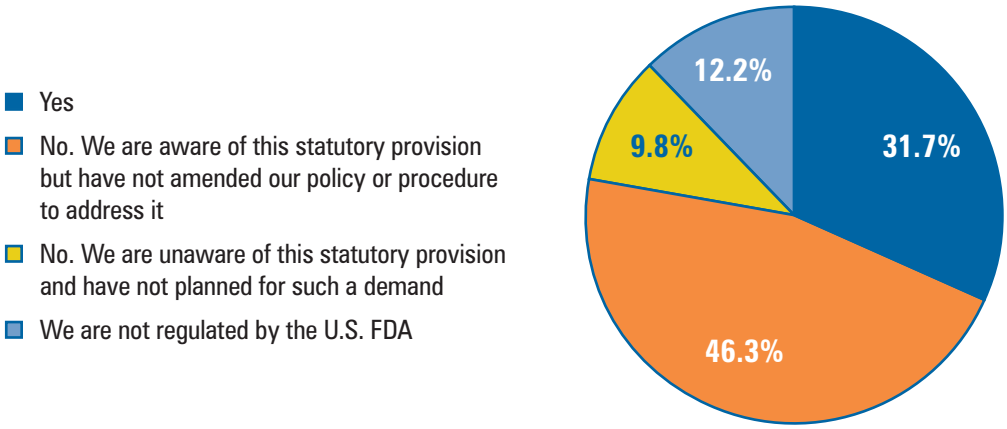
31. *Have you incorporated into your inspection management policy or procedures a means for dealing with a demand from the U.S. FDA for records "in advance of or in lieu of an inspection", issued pursuant to 21 USC 374(a)(4)?*

| Answer | Count | % |
|---|---|---|
| Yes. We have amended our policy or procedure to provide for the receipt of a demand for records from the U.S. FDA | 13 | 31.7% |
| No. We are aware of this statutory provision but have not amended our policy or procedure to address it | 19 | 46.3% |
| No. We are unaware of this statutory provision and have not planned for such a demand | 4 | 9.8% |
| We are not regulated by the U.S. FDA | 5 | 12.2% |
| ***Answered*** | ***41*** | |





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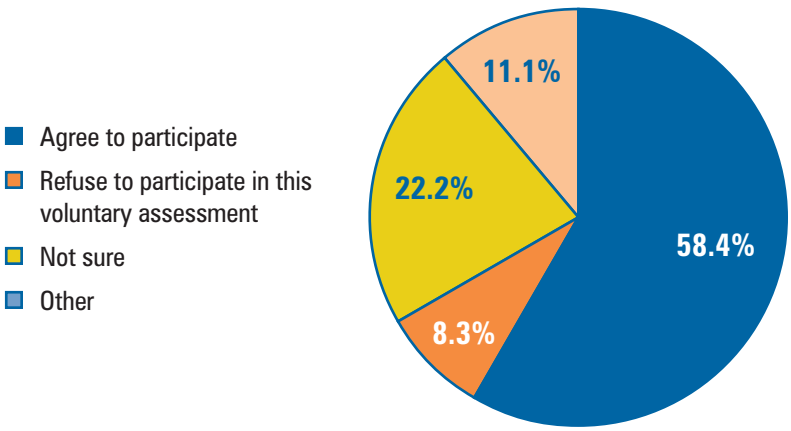
32. *If the U.S. FDA notified you that they want to conduct a "voluntary regulatory assessment" which would not technically be an inspection, but the results of which could determine approval or withholding of approval of a pending submission (NDA, ANDA, BLA, etc.), would you:*

| Answer | Count | % |
|---|---|---|
| Agree to participate | 21 | 58.4% |
| Refuse to participate in this voluntary assessment, but state that you would fully cooperate with an official inspection that met statutory requirements | 3 | 8.3% |
| Not sure | 8 | 22.2% |
| Other (Please specify) | 4 | 11.1% |
| ***Answered*** | ***36*** | |

Other responses:

- Decision based on business risk
- Might depend upon the site they want to inspect
- This is double work. The US REGULATIONS are allowing remote inspections; guidelines may be a limiting factor
- Would need to assess through legal




## Survey Conclusion

This survey will support PDA, manufacturers, and regulators in understanding the current state, the needs of manufacturers, and the challenges to be overcome in moving forward with remote regulatory inspections. The survey provides benchmarking information regarding experiences to date. In particular, the survey results illustrate the steps that companies have taken to prepare for and respond to remote regulatory inspections by global health authorities during the COVID-19 pandemic. These insights will continue to be useful because the use of remote inspection tools is expected to continue in some cases even after the pandemic is under control.

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